Review of MOS device and problems with classical scaling

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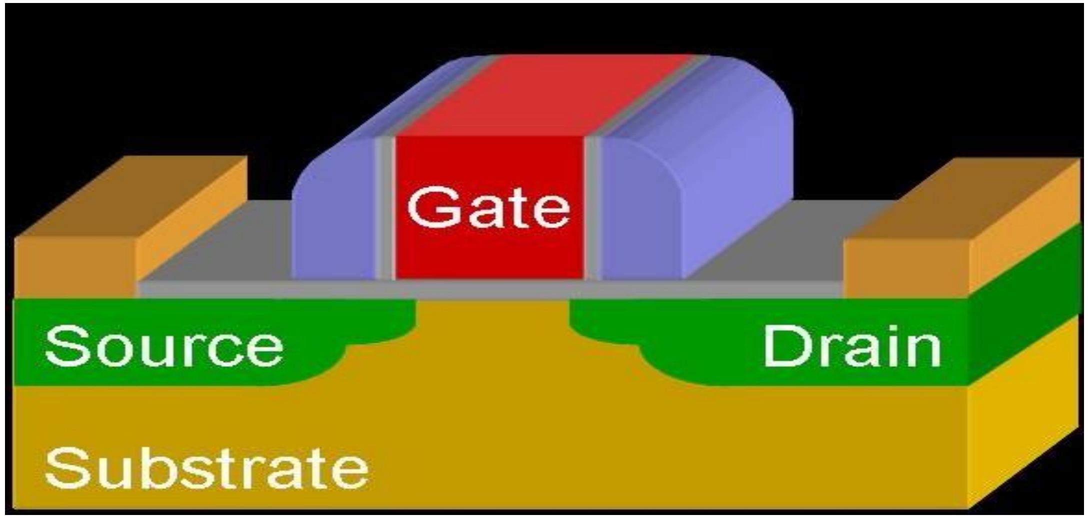
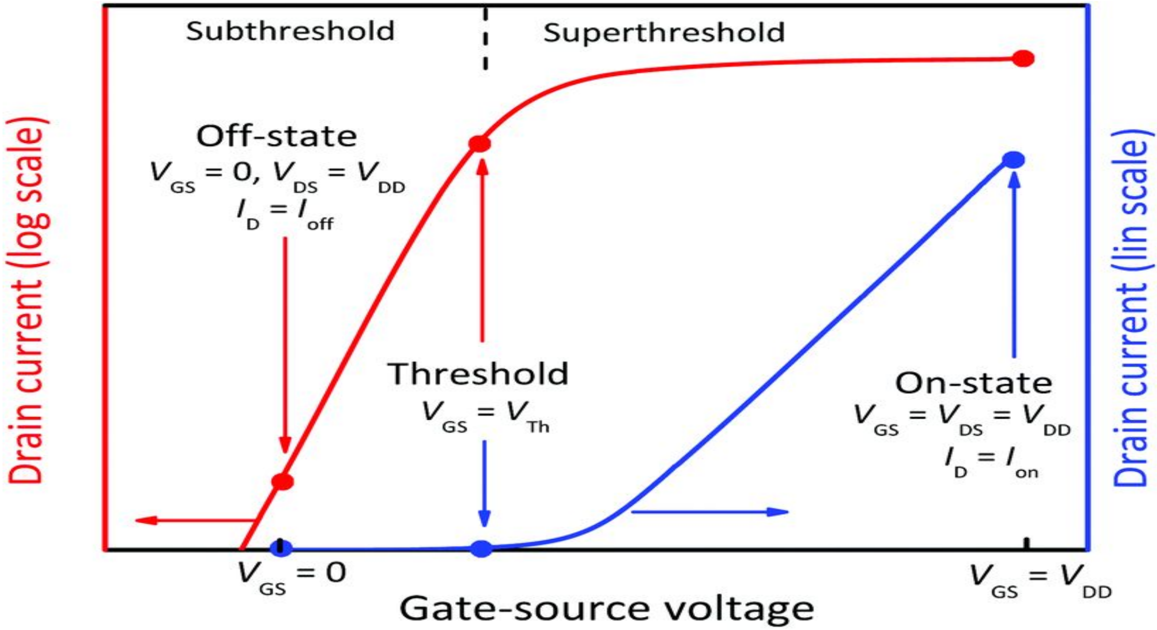
Course content

- Review of MOS transistor and problems with traditional (Dennard) scaling
- Review of MOS transistor technology
- Basic Quantum mechanics and Silicon band structure
- Mobility enhancement techniques (introduced in high volume manufacturing in 2003)
- High k gate dielectrics and metal gates (introduced in 2007)
- FinFETs (introduced in 2012)
- High mobility channel materials (introduced in 2011)
- Nanosheet transistors (introduced in 2024 Samsung Galaxy Watch7)
- Layout dependent effects (became a serious issue since late 2000s)

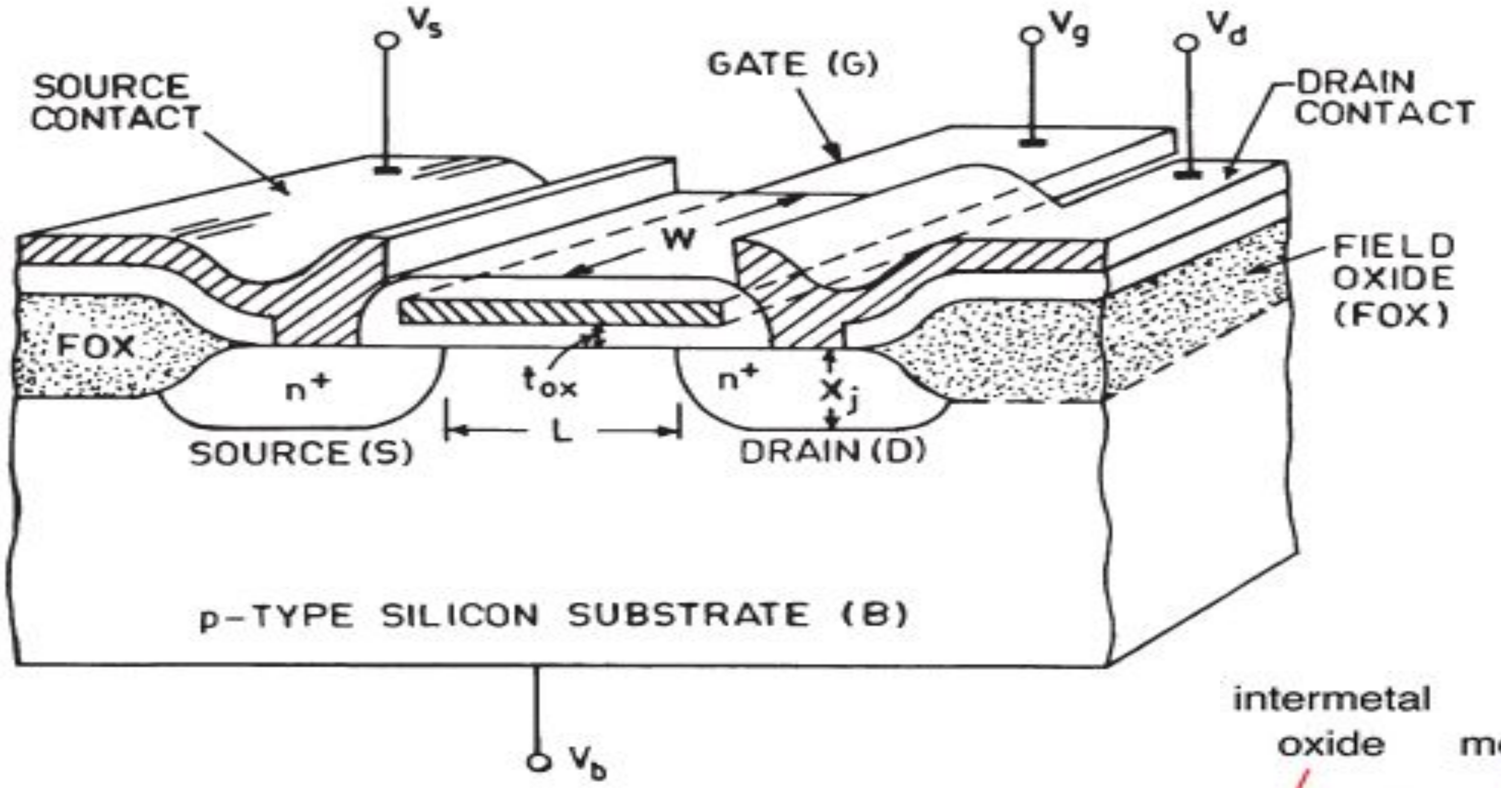
MOS Field Effect Transistor

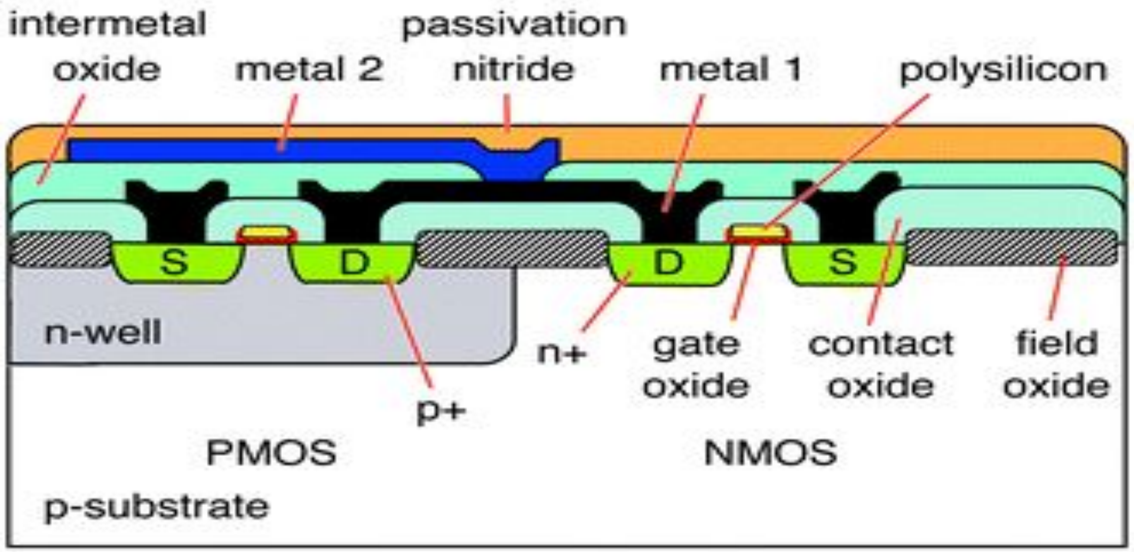
- The MOSFET is a MOS capacitor with Source/Drain terminals
- How does it work?
  - Gate voltage (V_{GS}) controls mobile charge under gate
  - Source-drain voltage (V_{DS}) sweeps this charge creating current (I_D)
- Desired characteristics
  - High “On” current (300-1300 μA/μm)
  - Low “Off” current (10 pA-100 nA/μm)





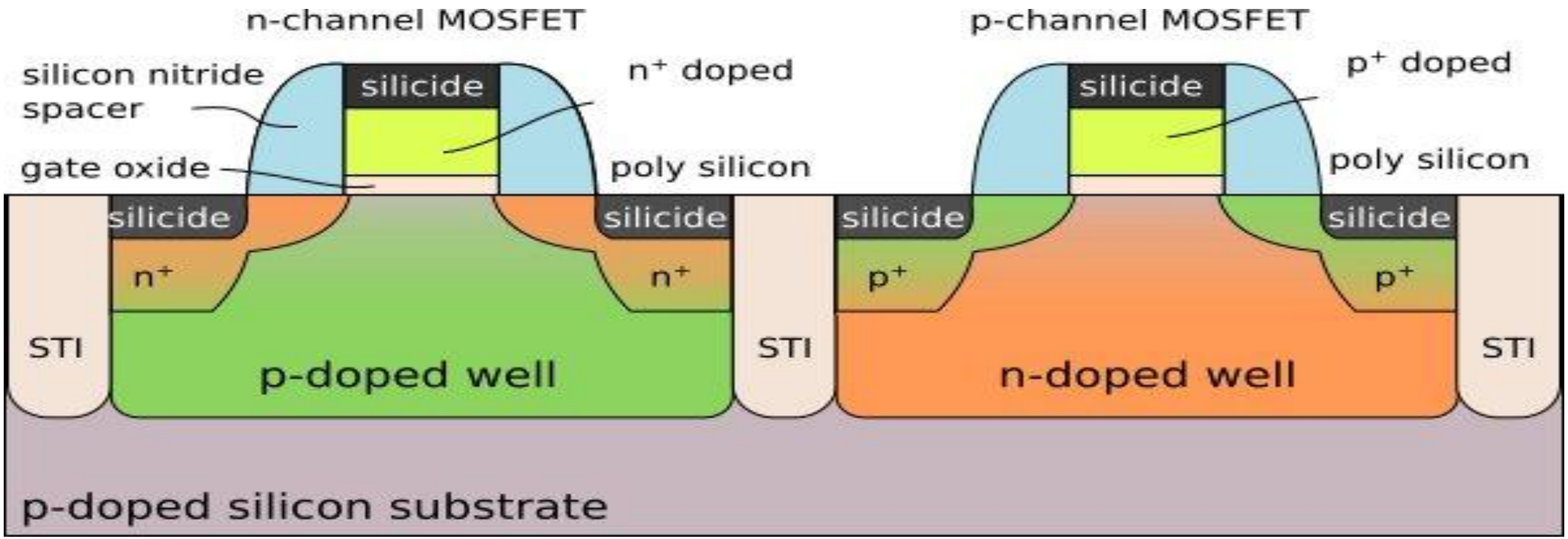
MOS Field Effect Transistor (1980s)





Single well CMOS process (1-2 µm)

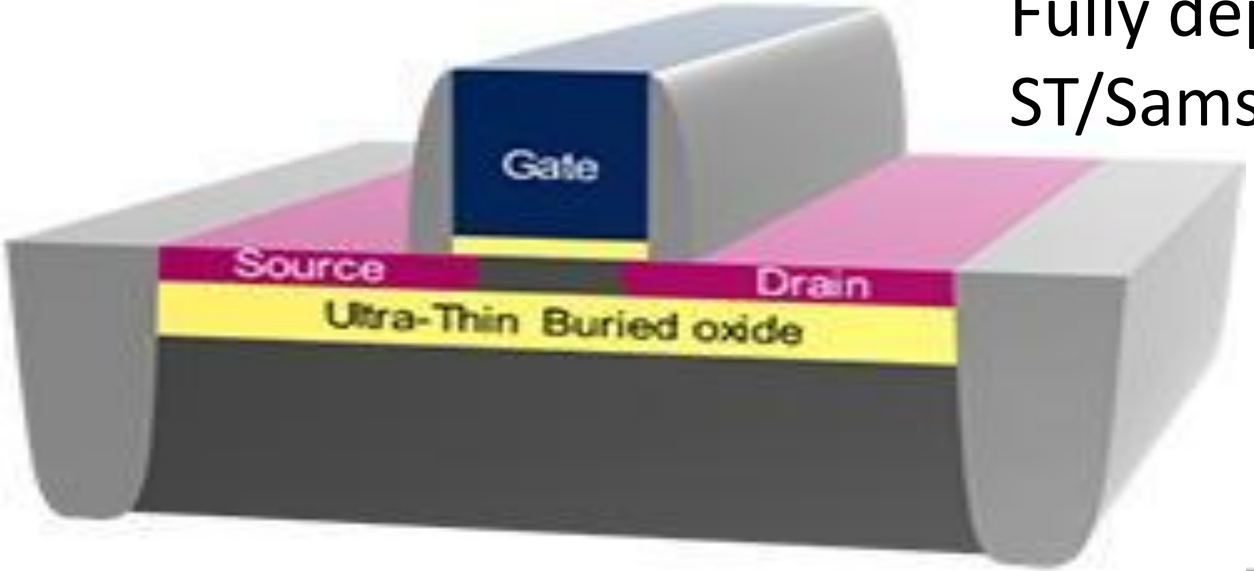
MOS Field Effect Transistor (1990s-2003)

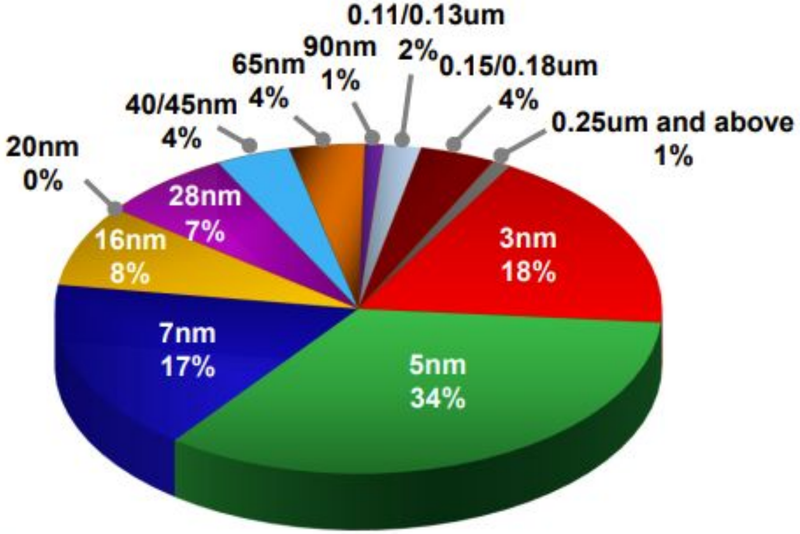


Twin-well CMOS process (< 1 µm)

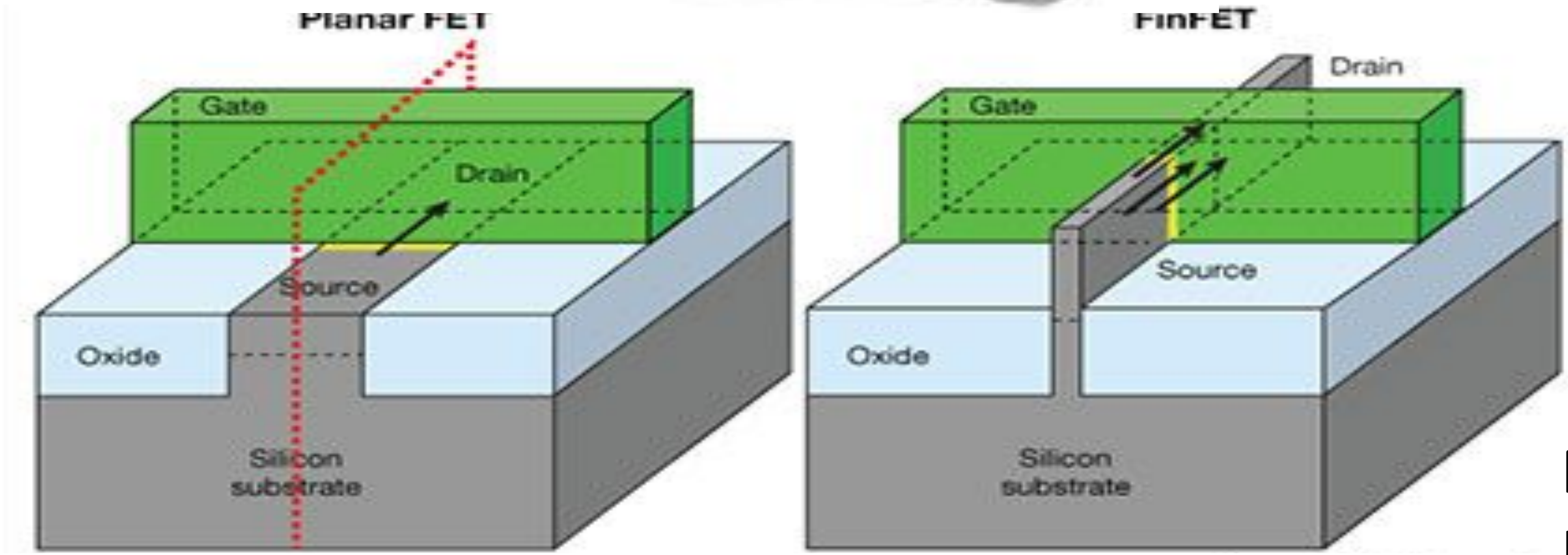
Modern day MOSFET (two variants)

Fully depleted Silicon on Insulator (FDSOI)
ST/Samsung/GF/NXP



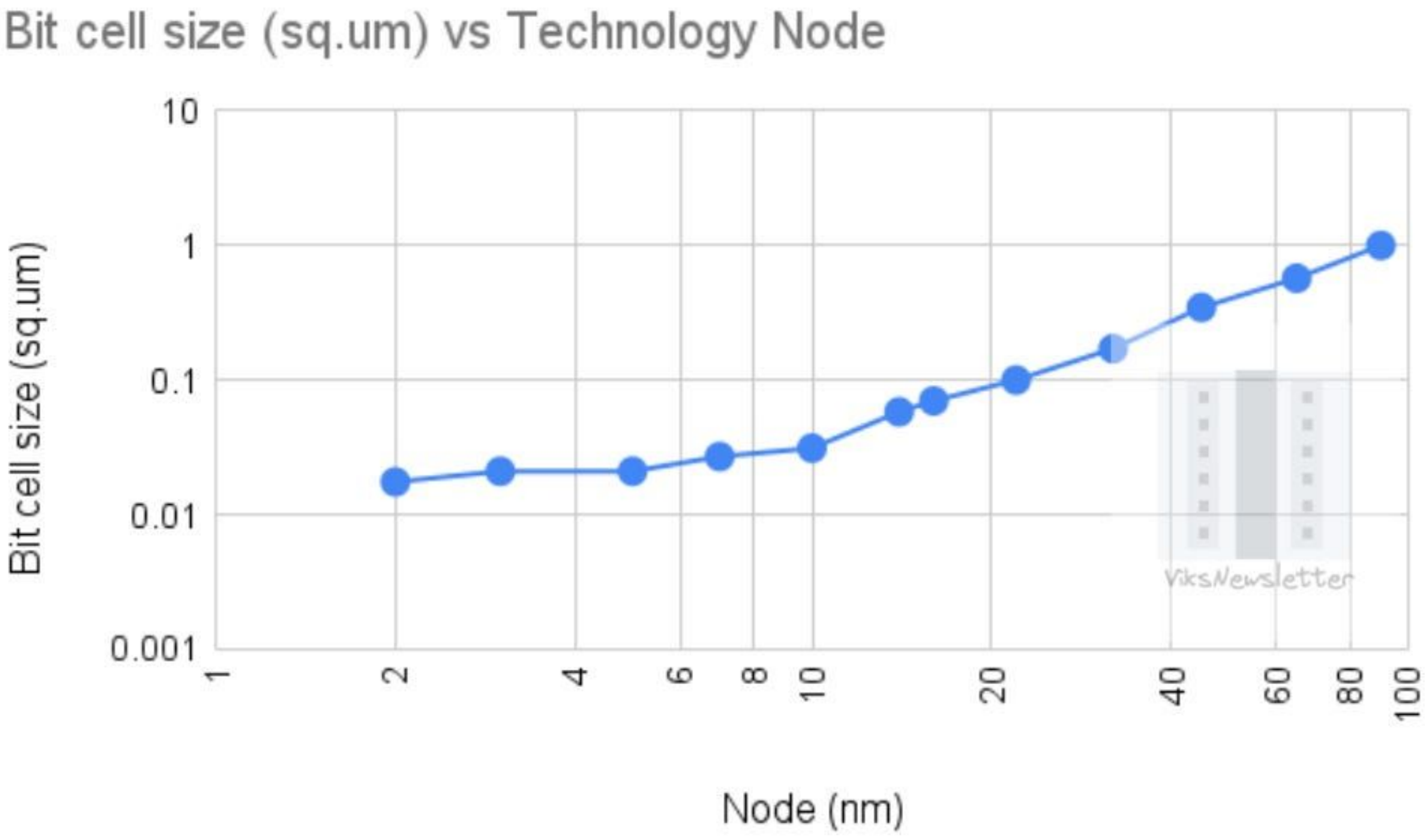


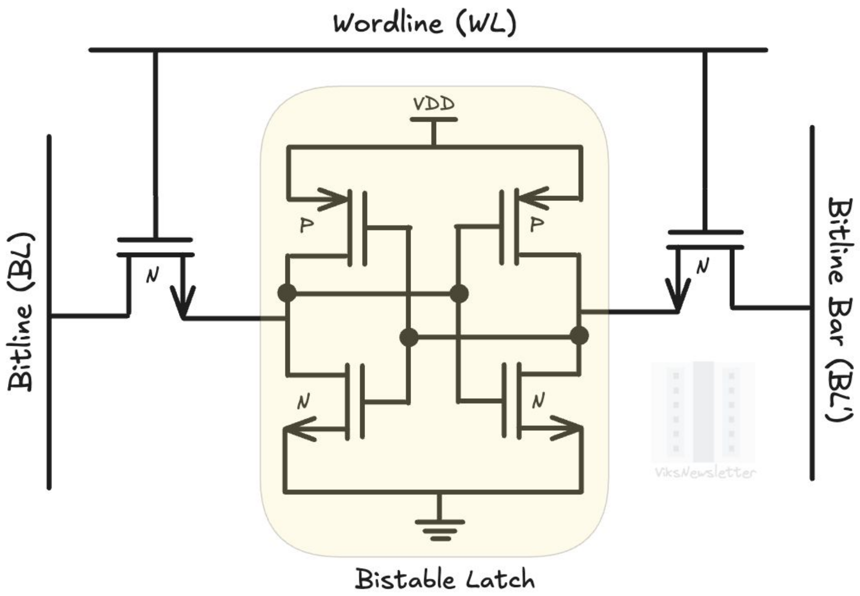
TSMC 2024 revenue



FinFET (16-3nm)
Intel/Samsung/GF/TSMC

FinFET scaling is over?

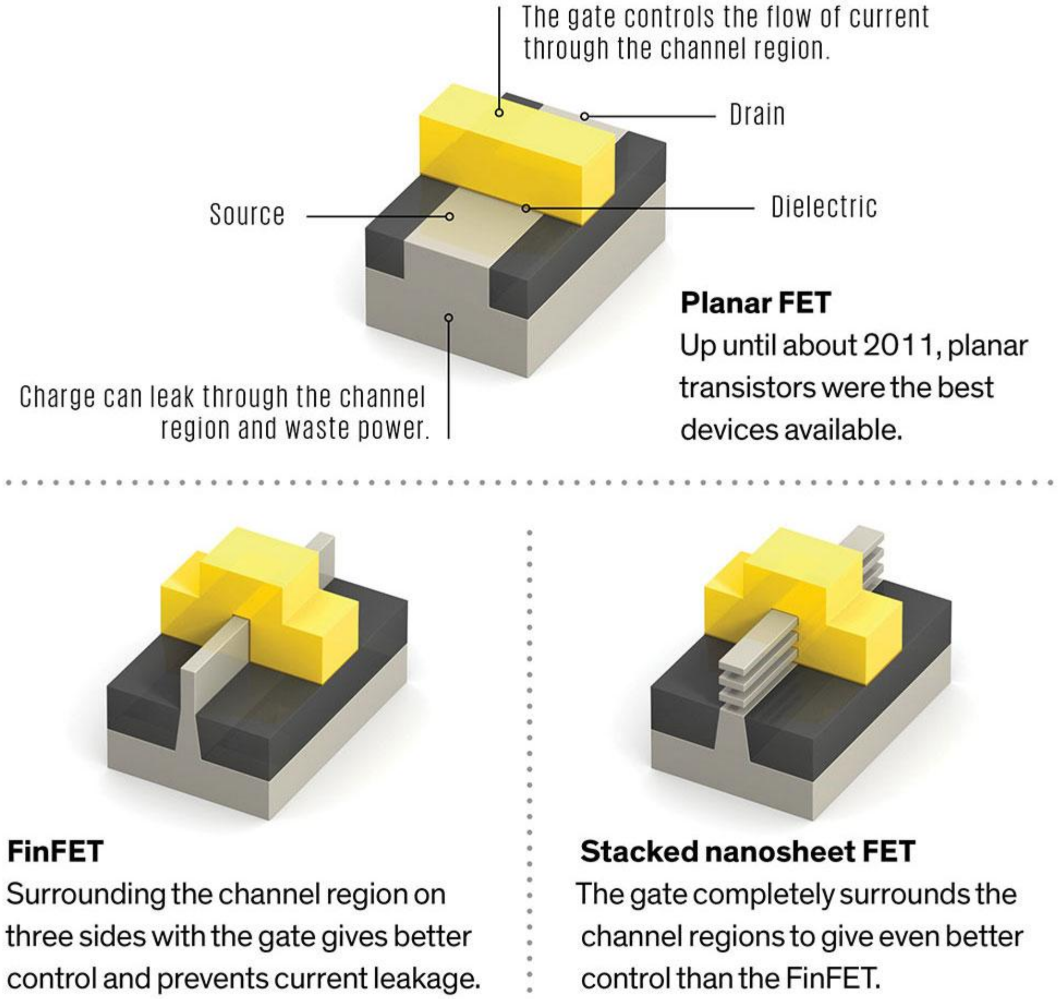


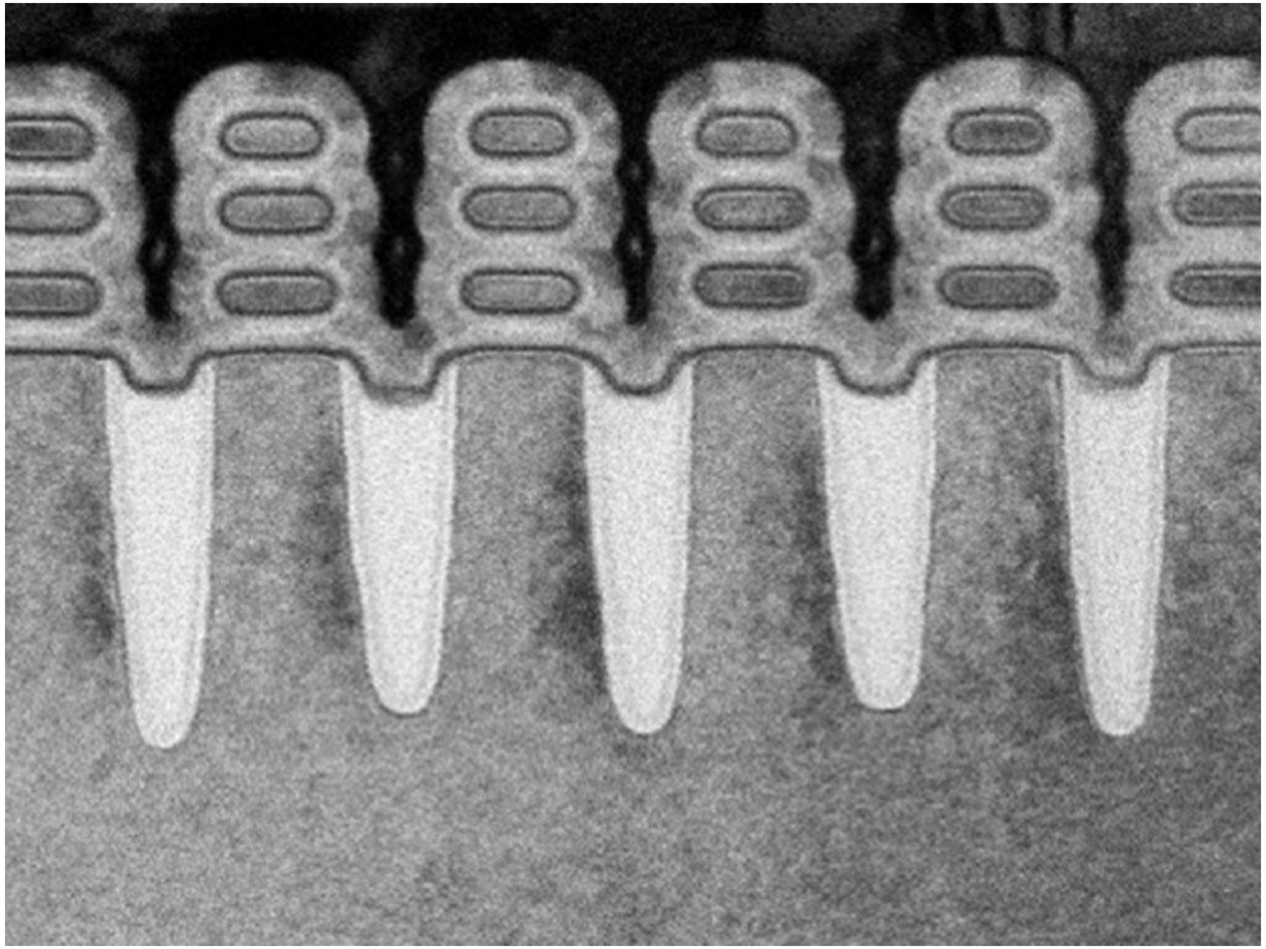


Around the 5-7nm node, SRAM scaling seemed to flatten out

https://www.viksnewsletter.com/p/a-close-look-at-sram-for-inference

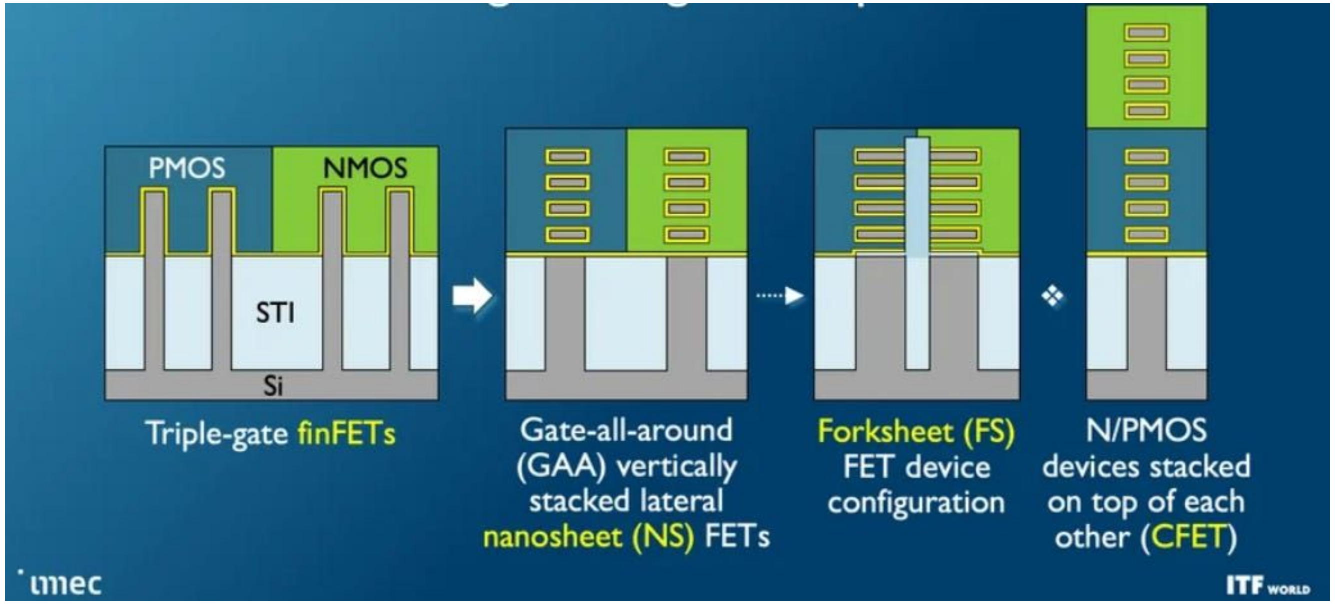
Most recent form of MOSFET- Nanosheet transistor

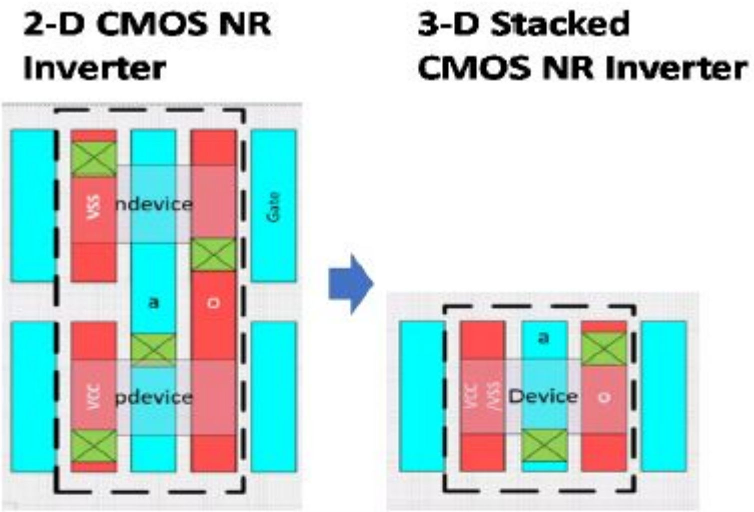


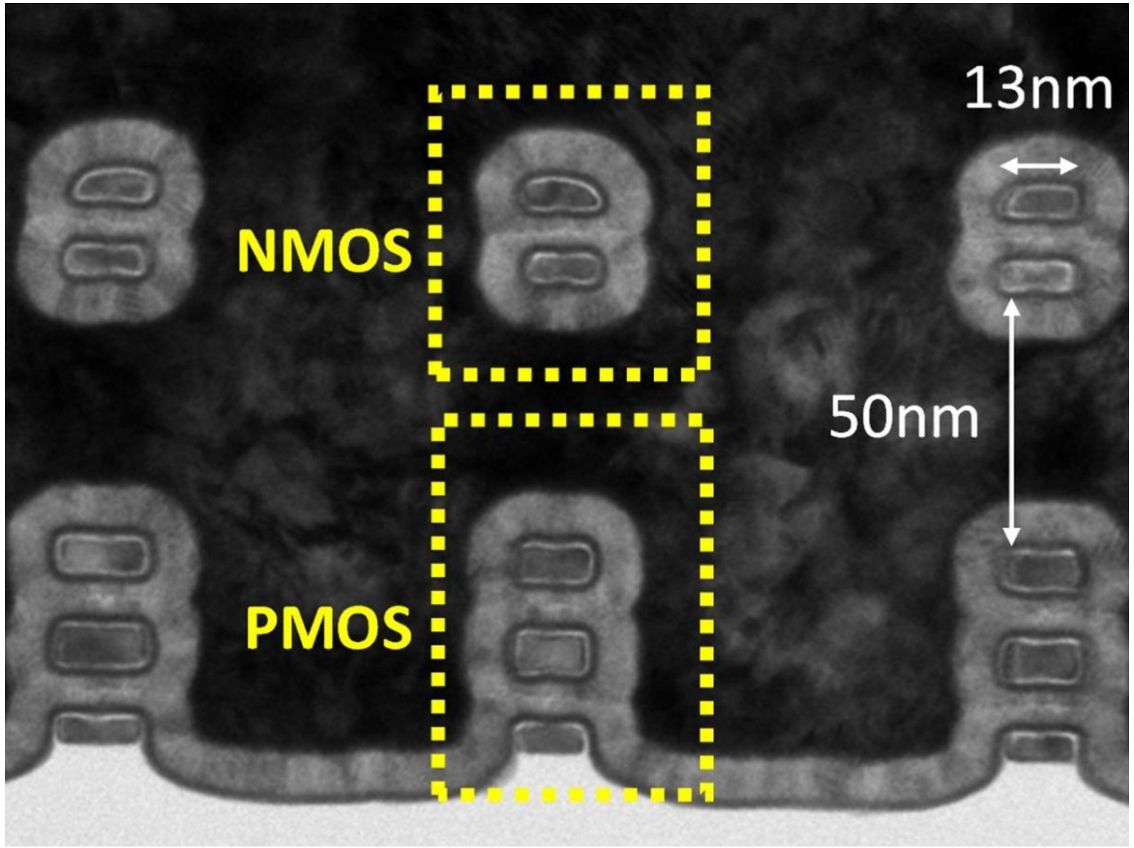


Samsung VLSI 2004 (Multi-bridge channel FETs)
https://spectrum.ieee.org/semiconductors/devices/the-nanosheet-transistor-is-the-next-and-maybe-last-step-in-moores-law

Future MOSFET- Stacked Nanosheet transistor or CFET







https://spectrum.ieee.org/nanoclast/semiconductors/devices/intels-stacked-nanosheet-transistors-could-be-the-next-step-in-moores-law

MOSFET Scaling

- MOSFETs have been steadily miniaturized over time
  - 1970s: Gate length ~ 10 μm
  - Today: Gate length ~ 15 nm

  ~ 15 generations of MOSFET with minor/major modifications over previous generations

- Why they look different ? - Process and Device changes are required to improve (or not worsen) performance /power /reliability /manufacturability of the devices from one generation to other

- Reasons:
  - Improved circuit operating speed (~30% every gen)
  - Increased device density --> lower cost per function (~50% area reduction)
  - More functionality in device/gadget

Herbert Kroemer in his Nobel prize lecture

Whenever I teach my semiconductor device physics course, one of the central messages I try to get across early is the importance of energy band diagrams. I often put this in the form of "Kroemer's Lemma of Proven Ignorance":

> If, in discussing a semiconductor problem, you cannot draw an **Energy Band Diagram**, this shows that you don't know what you are talking about,

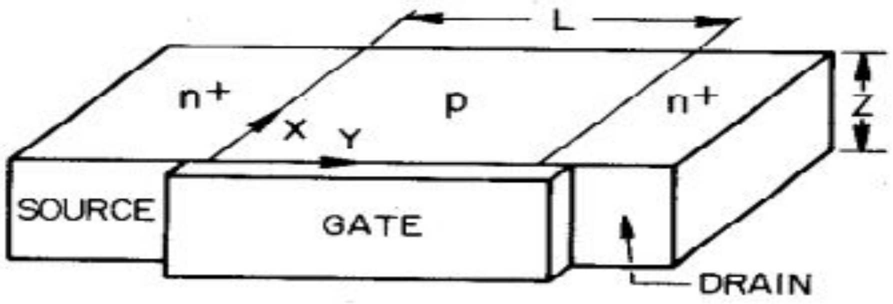
with the corollary

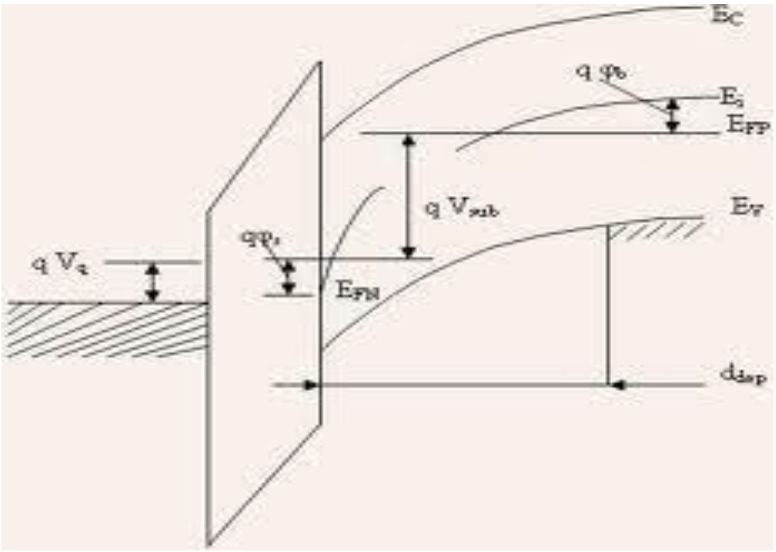
> If you can draw one, but don't, then your audience won't know what you are talking about.

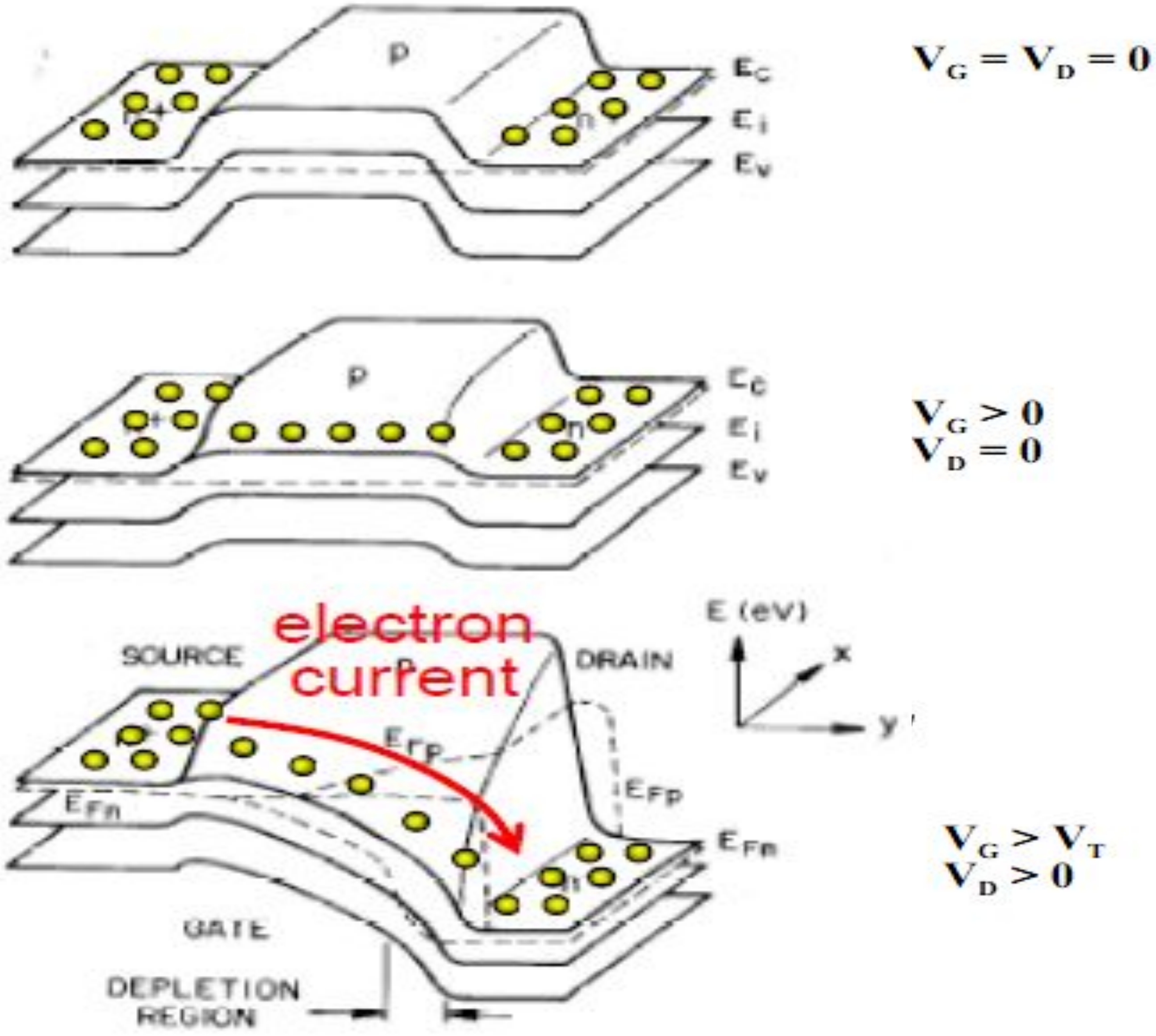
Nowhere is this more true than in the discussion of heterostructures, and much of the understanding of the latter is based on one's ability to draw their band diagrams – and knowing what they mean.

https://www.nobelprize.org/uploads/2018/06/kroemer-lecture.pdf

MOSFET 2-D band diagram

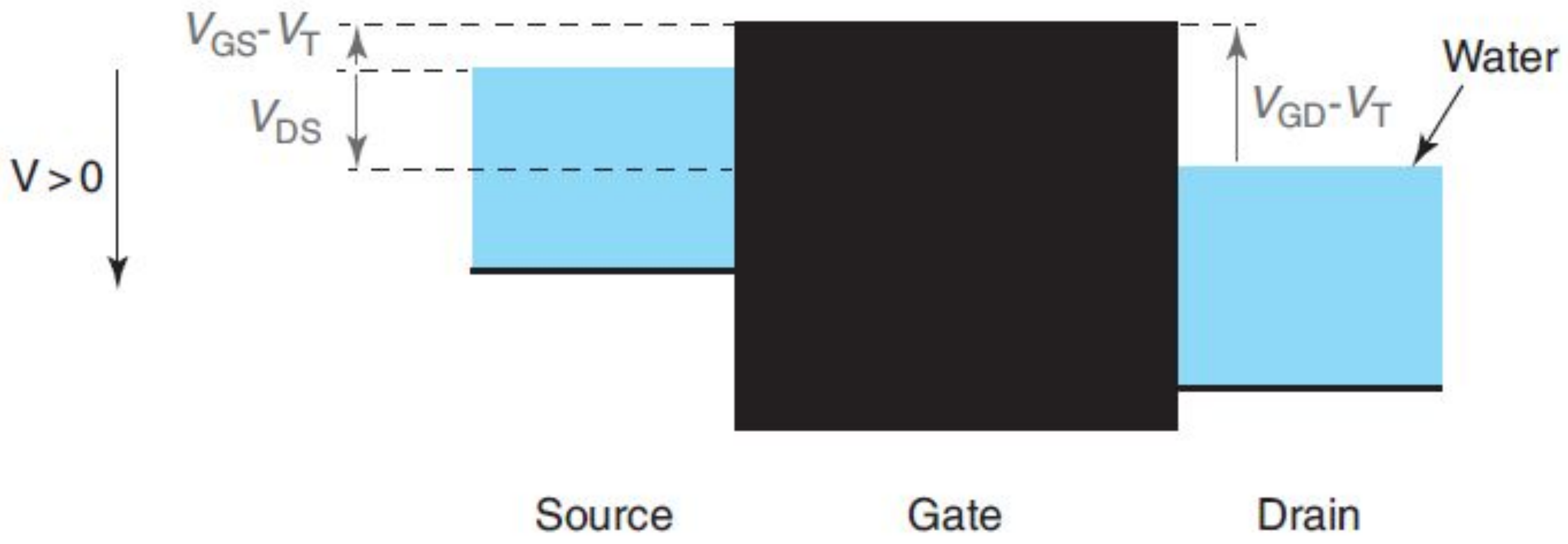






Gate voltage reduces the barrier from source/drain to channel enabling carrier injection

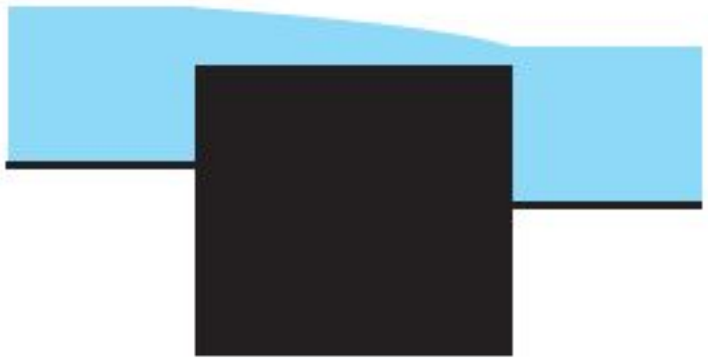
Qualitative operation of ideal MOSFET



The relative height of the two reservoirs of water corresponds to the drain-to-source voltage. The relative height of the barrier above the water level of the source corresponds to the gate to source voltage in excess of threshold. At a value of V_{GS} = V_T, the top of the barrier is leveled with the water surface.

Reservoir analogy of an ideal MOSFET

Cut-off

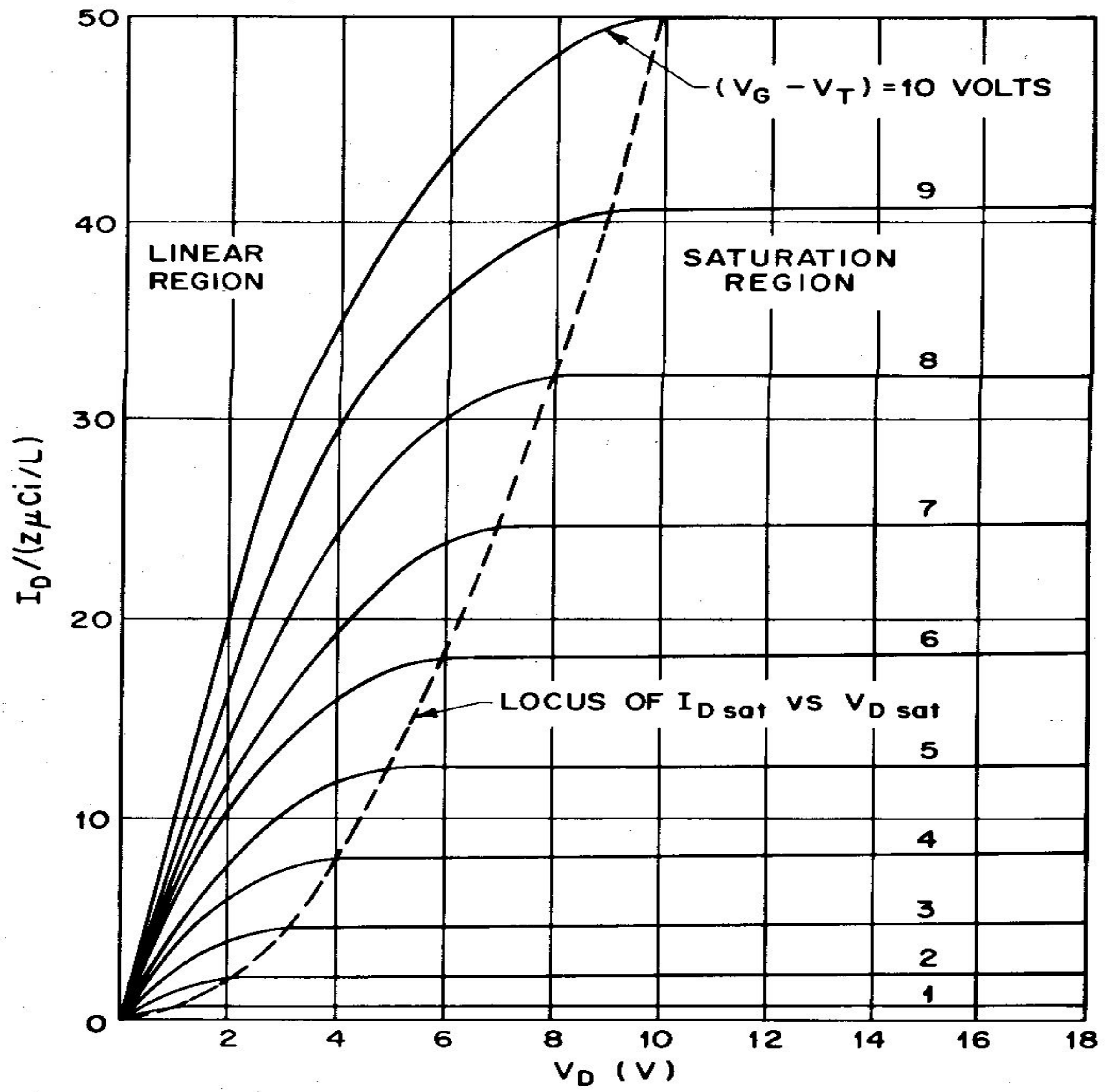


Linear or triode

Saturation

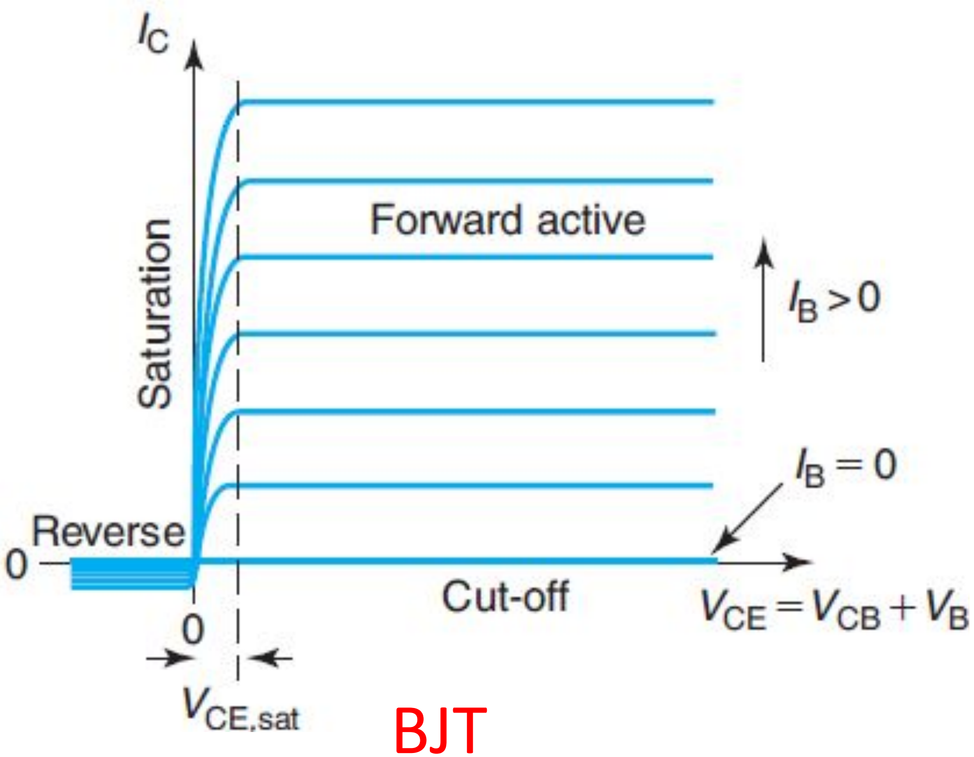
In the cut-off regime, the barrier is above the water level on the source and drain and water does not flow. In the triode or linear regime, the barrier is below the water level of both the source and drain. Water flows from source to drain. In the saturation regime, the barrier is below the water level of the source but above that of the drain. Water flows from source to drain, but the water flow rate is independent of the relative levels of the source and drain.

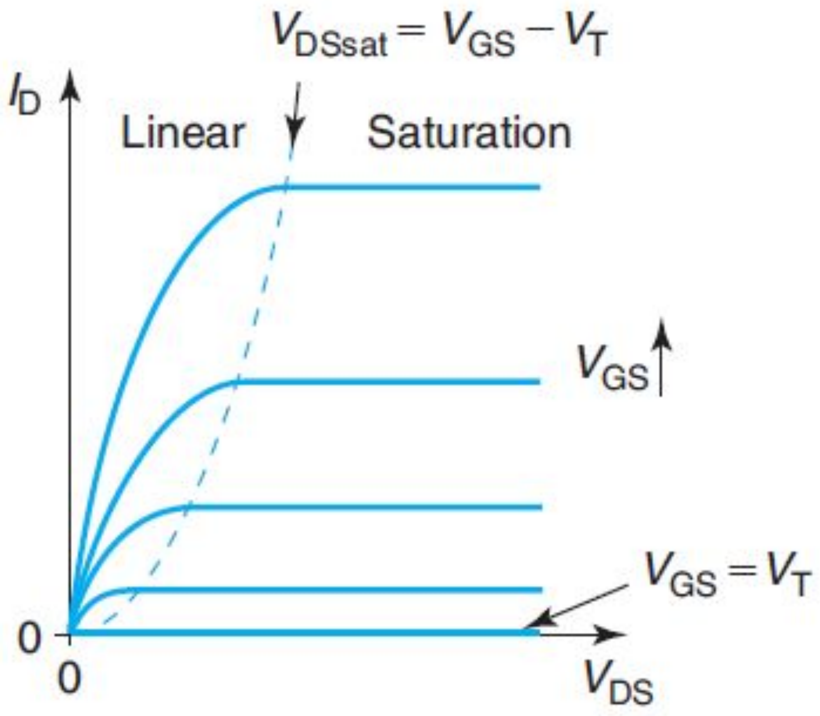
Ideal MOSFET *I-V* Characteristics



Enhancement-Mode N-channel MOSFET

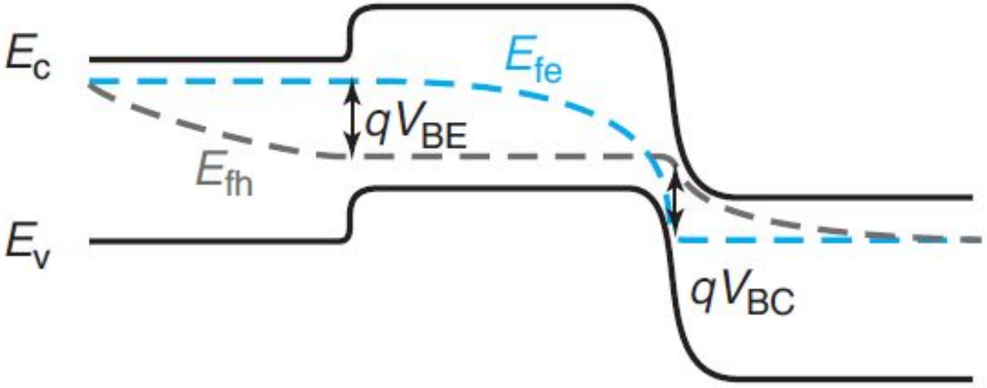
Output characteristics - Comparing BJT & MOSFET

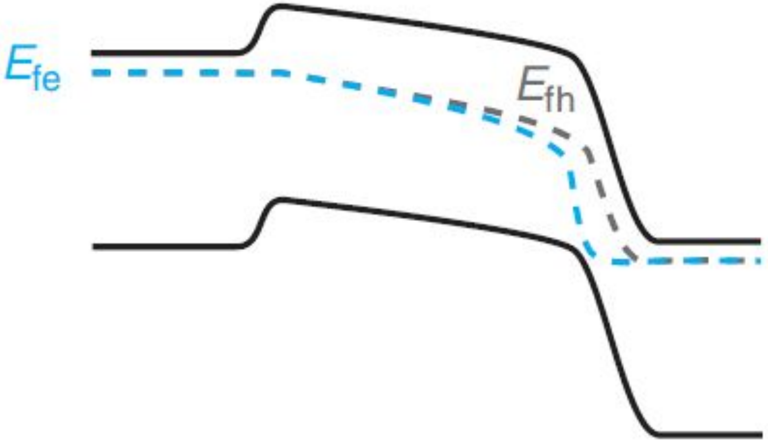




BJT

MOSFET





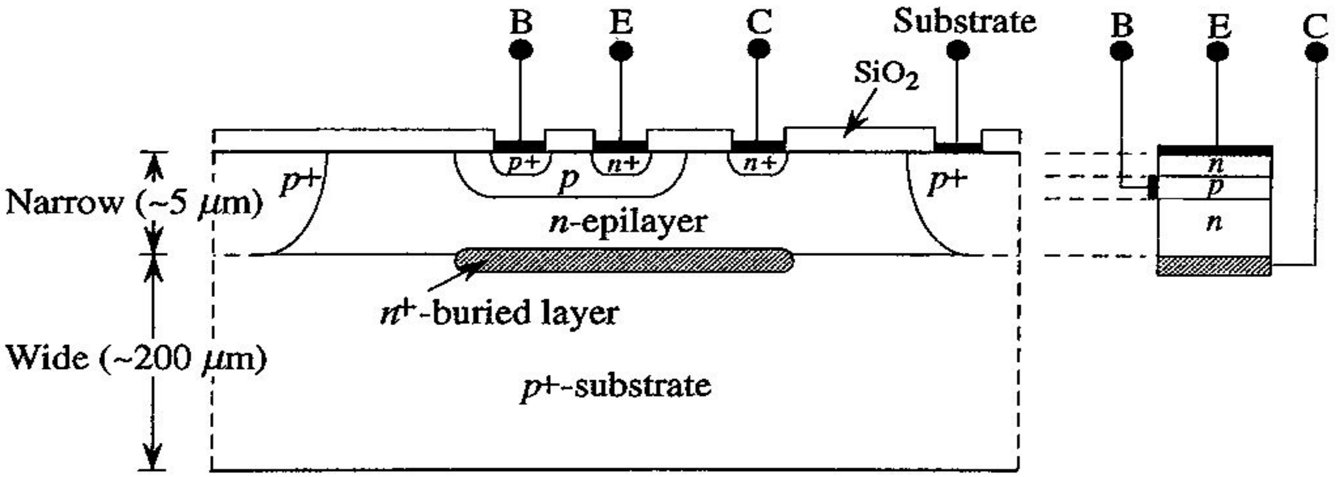
Modulate injection of carriers to base or channel

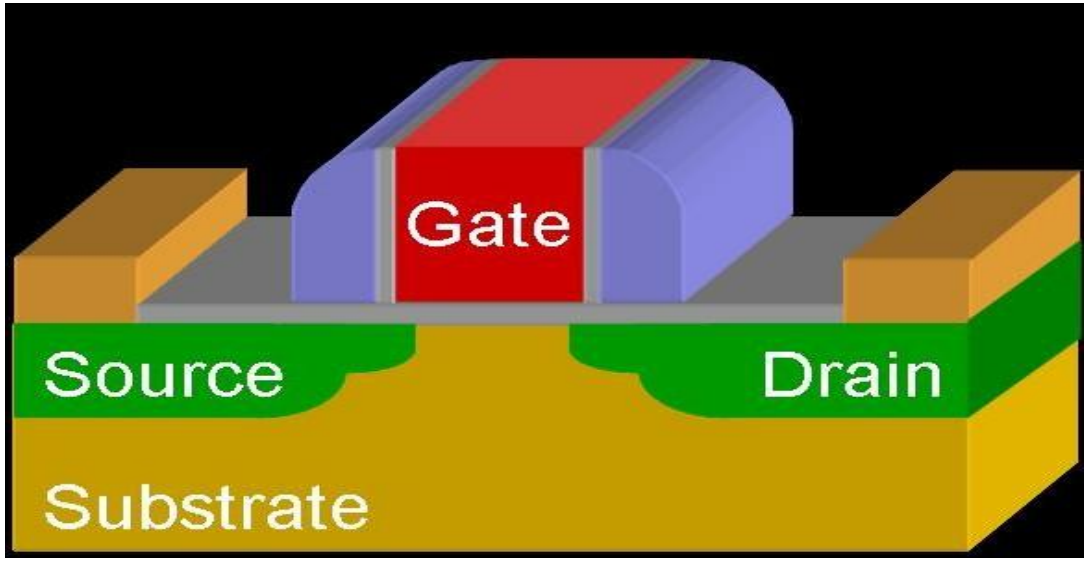
Collect the carriers at a reverse biased junction

Transport is diffusion in base and drift in channel

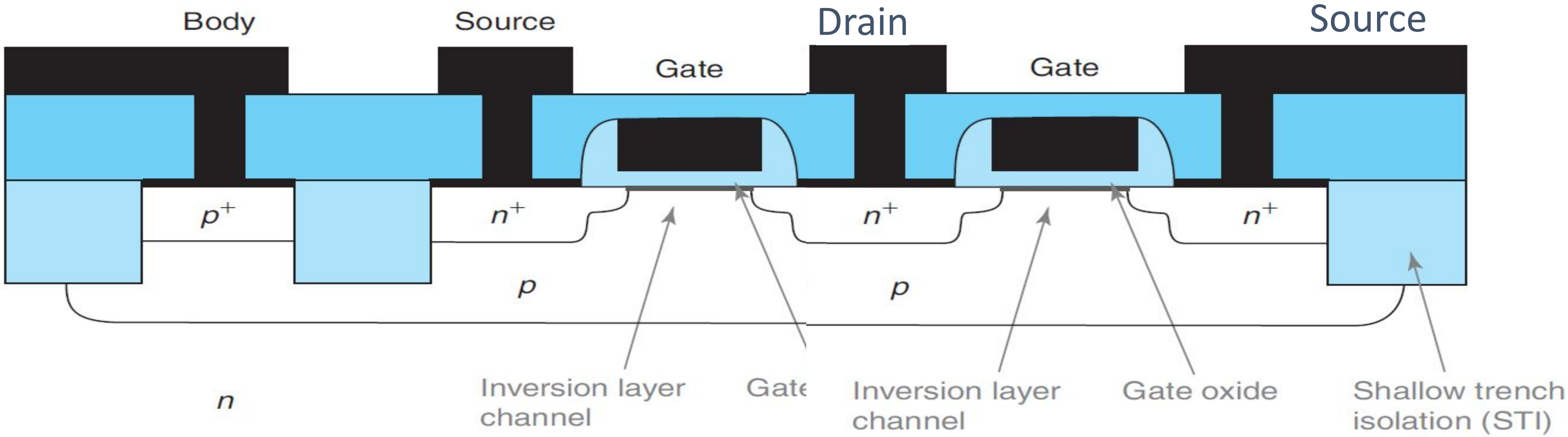
MOSFET Scaling

- Advantages over BJT:
  - Critical dimension in lateral dimension compared vertical in BJT. Possible to make transistors with different gate lengths on the same wafer.
  - No bipolar logic family with very low standby power dissipation as CMOS. Limits no: of transistors in a chip.
  - MOS is a symmetrical device compared to BJT. More flexibility in design.



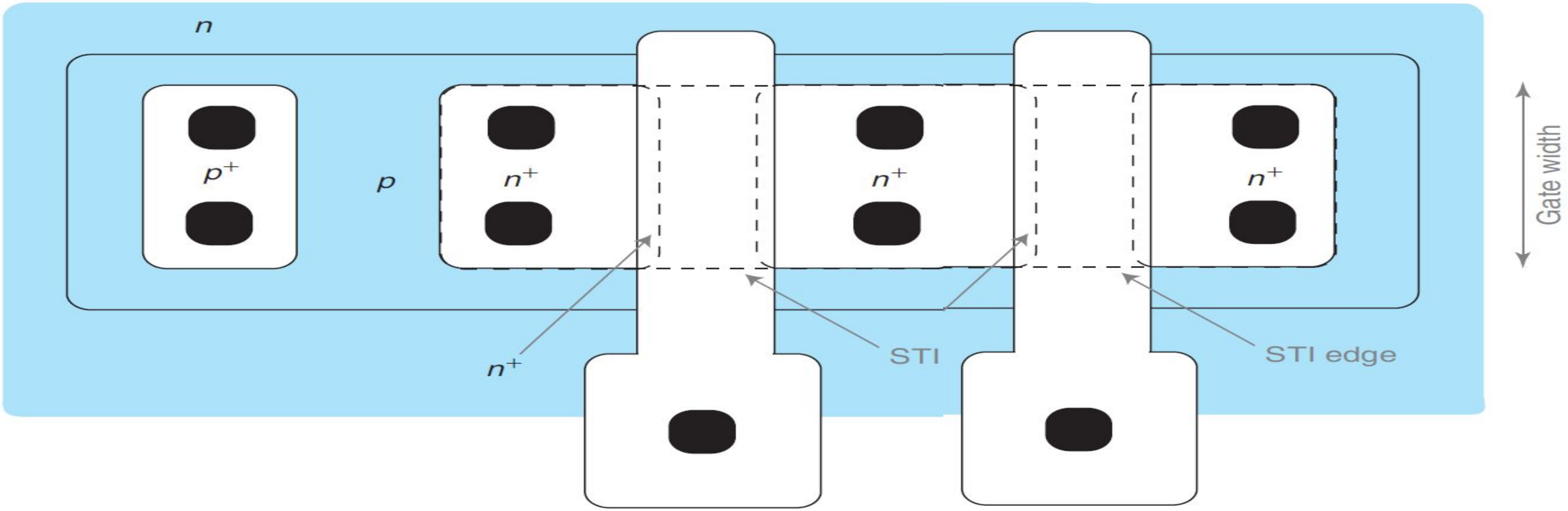


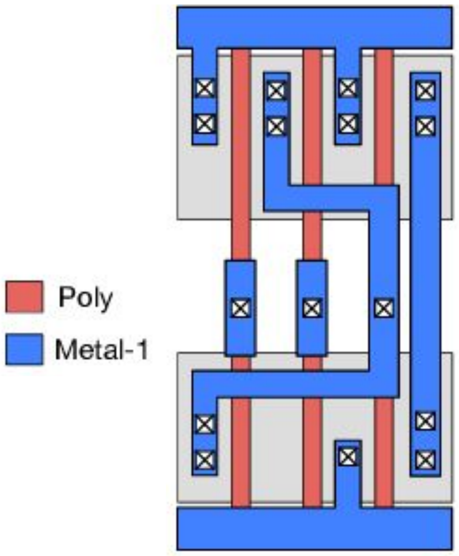
What does scaling means?



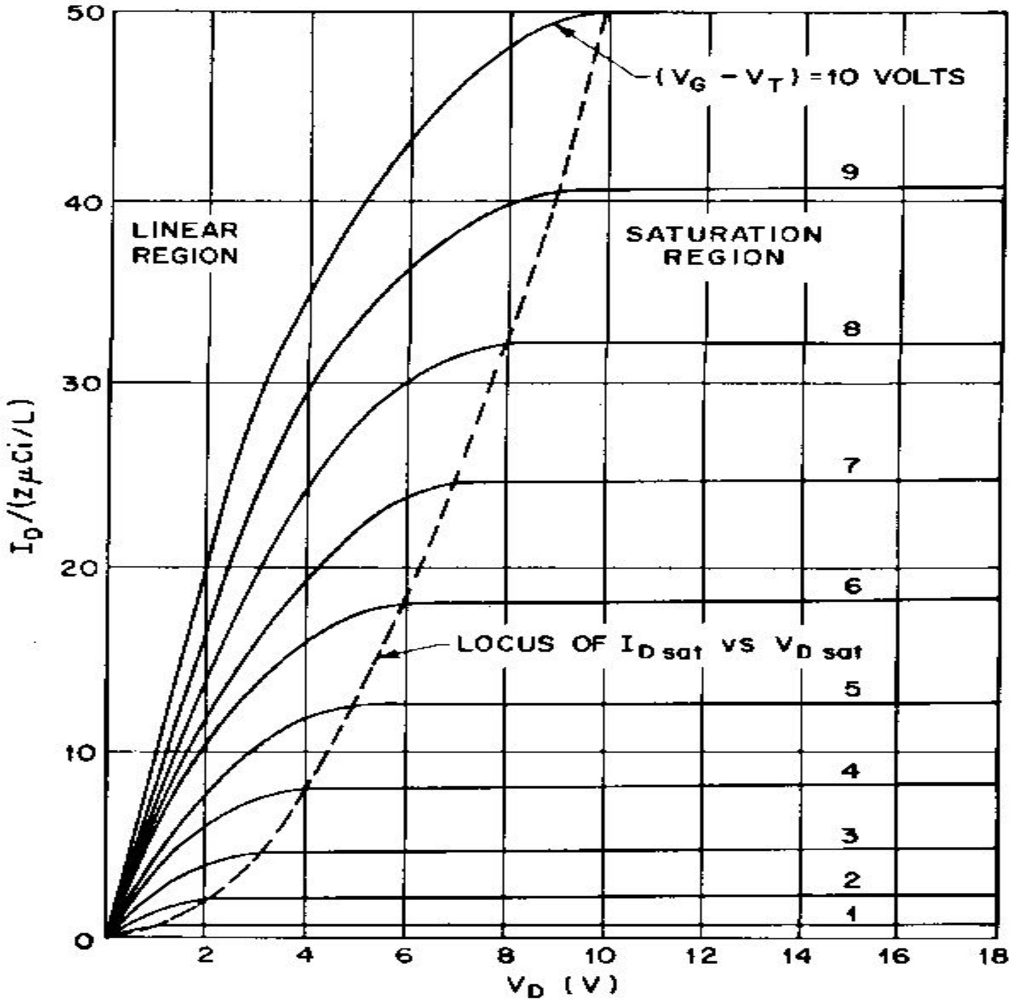
Reduce contacted gate pitch and metal wiring pitch

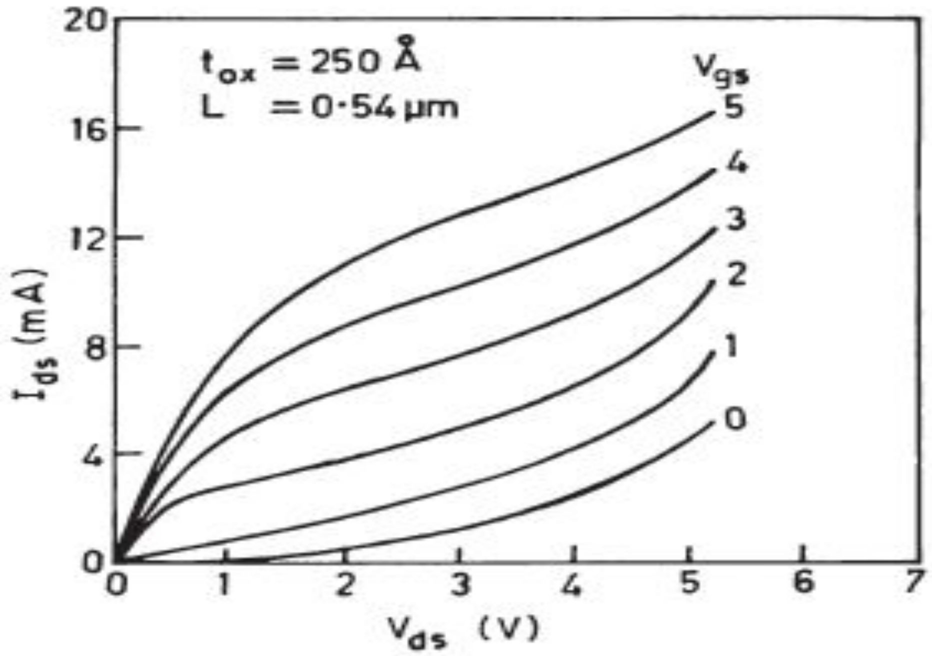
Contacted Gate pitch





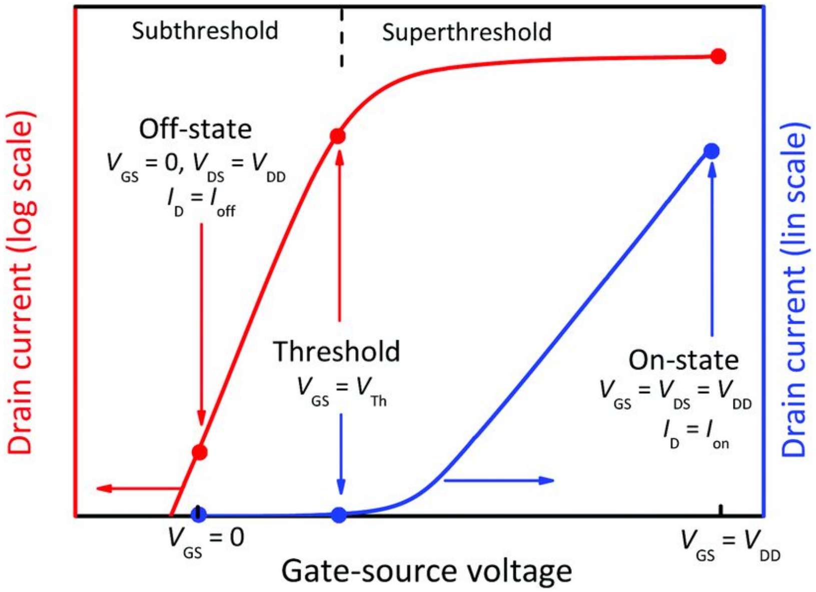
Bad scaling



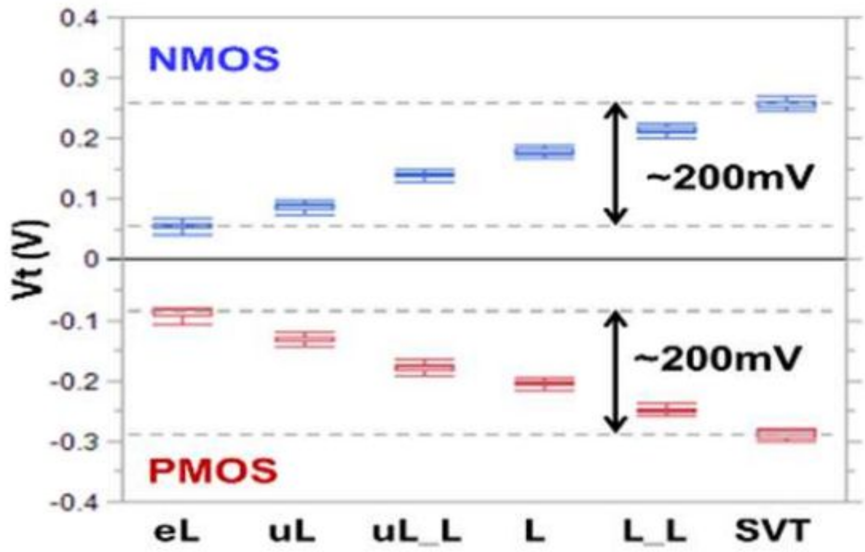


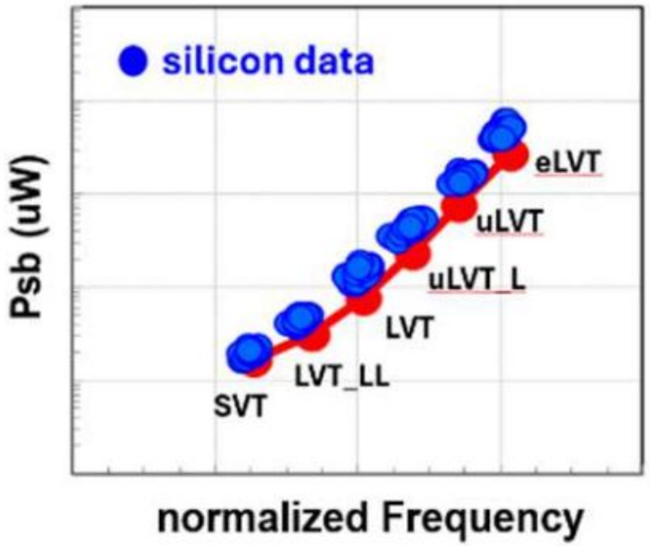
Robert Pierret, Semiconductor Device Fundamentals, 2005

Threshold voltage



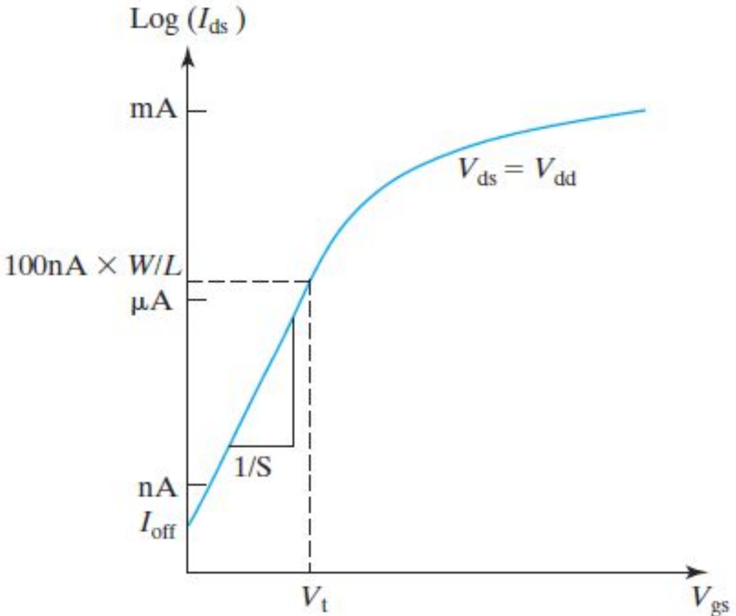
TSMC N2 from IEDM 2024





V_T is usually measured in industry using constant current method.

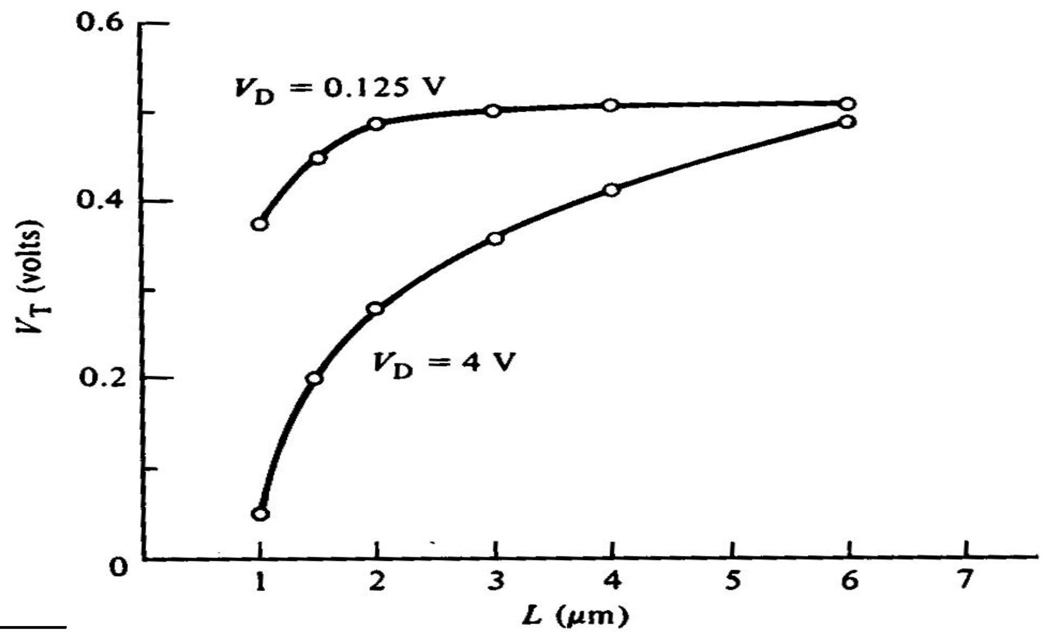
Foundry offers 4-6 different V_T flavors for power-performance trade-off



Short Channel Effect (SCE)

"V_T roll-off"

- $|V_T|$ decreases with L
  - Effect is exacerbated by high values of $|V_{DS}|$



$$V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2qN_A\varepsilon_{Si}(2\phi_F + V_{SB})}}{C_{ox}}$$

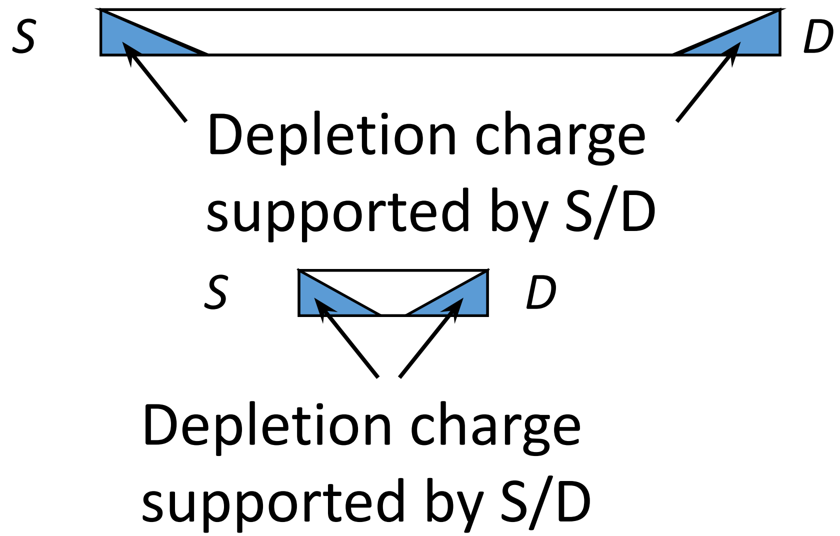
- This effect is undesirable (*i.e.* we want to minimize it!) because circuit designers would like V_T to be invariant with transistor dimensions and bias condition

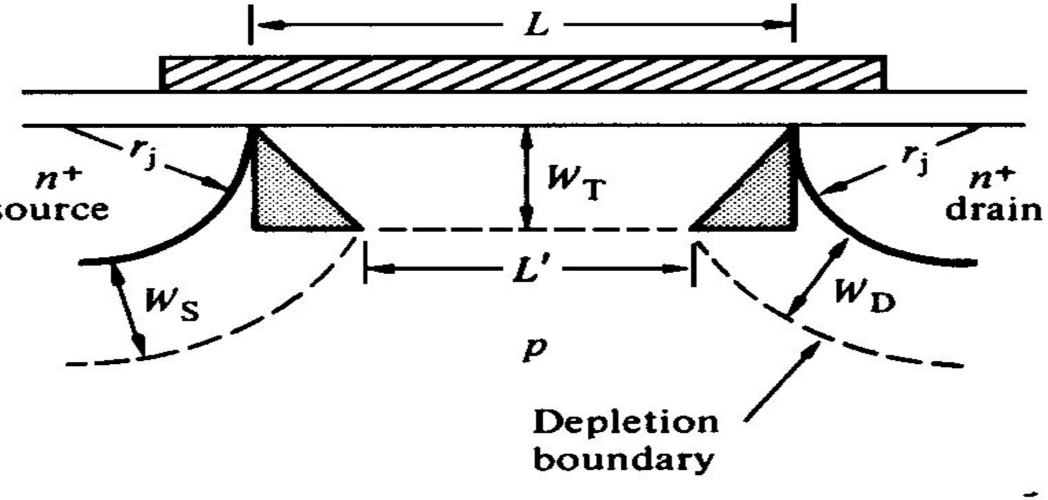
Qualitative Explanation of V_T roll-off

- Before an inversion layer forms beneath the gate, the surface of the Si underneath the gate must be depleted (to a depth W_T)
- The source & drain pn junctions assist in depleting the Si underneath the gate
  - Less gate charge is required to invert the semiconductor surface (*i.e.* $|V_T|$ decreases)

<u>Long *L*</u>:

<u>Short *L*</u>:

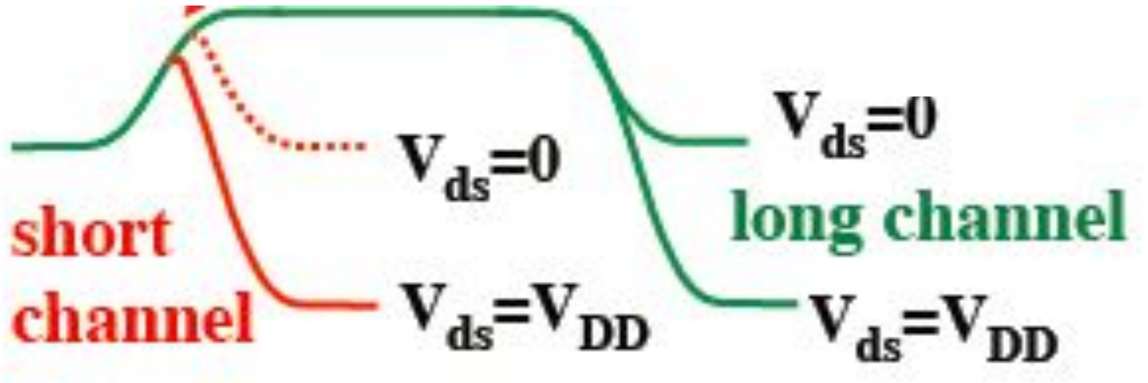


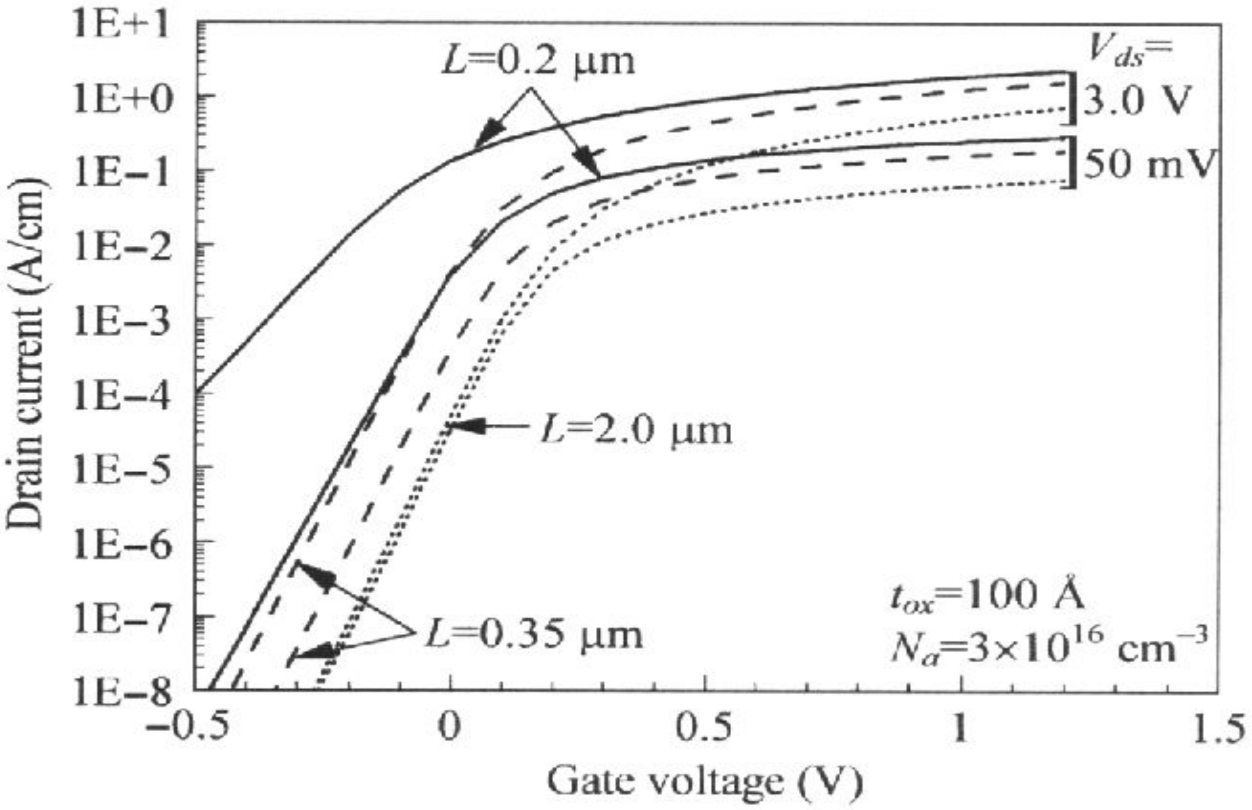


Drain Induced Barrier Lowering (DIBL)

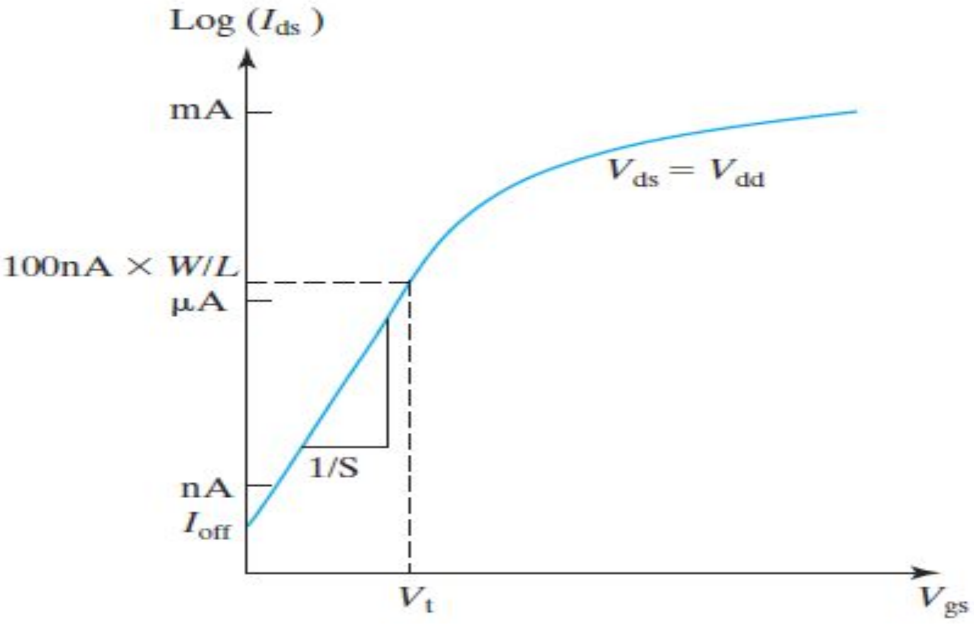
- As the source and drain get closer, they become electrostatically coupled, so that the drain bias can affect the potential barrier to carrier diffusion at the source junction

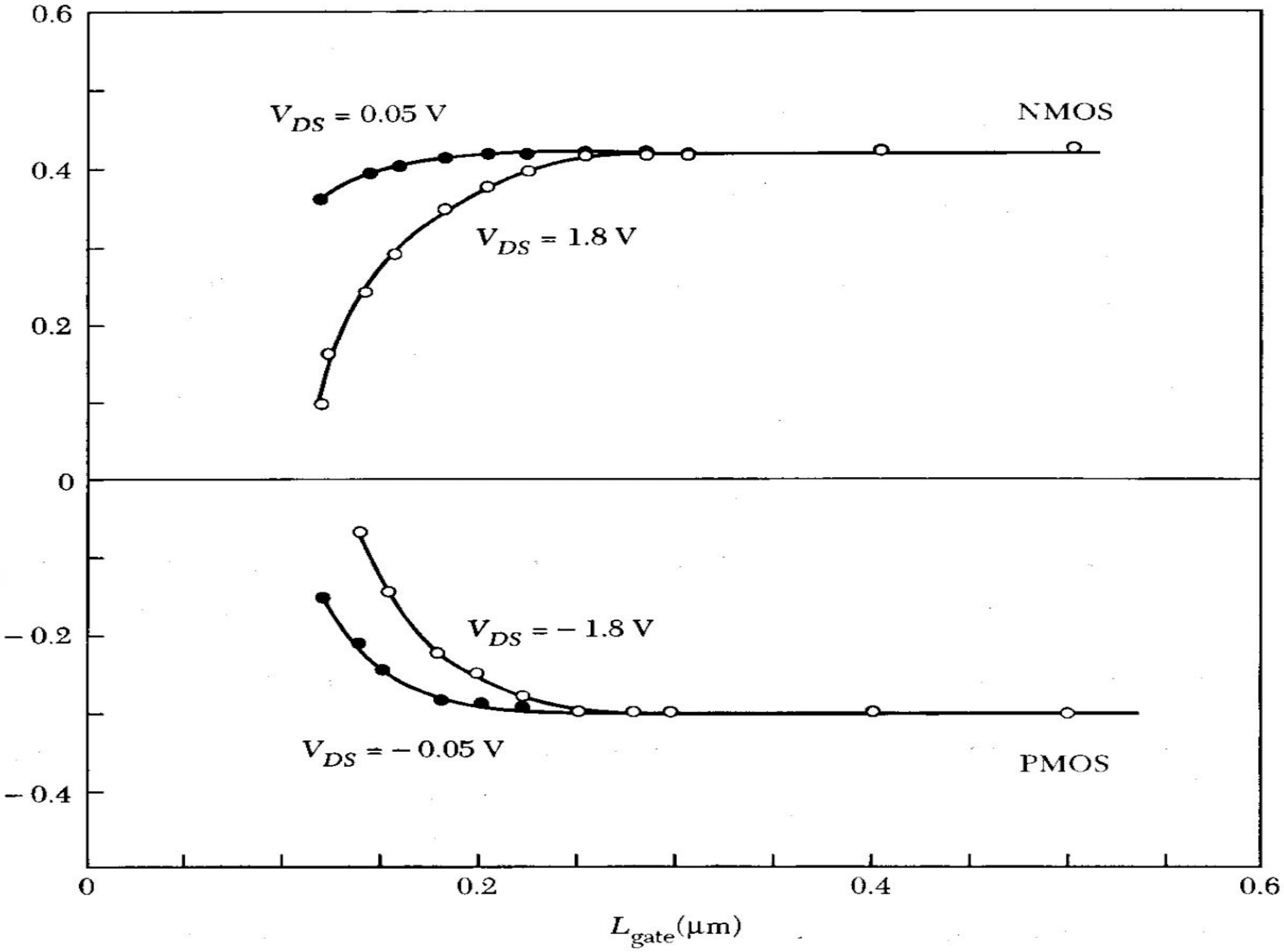
V_T decreases (*i.e.* OFF state leakage current increases)





V_T becomes a f(V_{DS}) for short channel devices.





V_T is usually measured in industry using constant current method

DIBL = $(V_{T,lin}-V_{T,sat})/(V_{DD}-V_{DS,lin})$

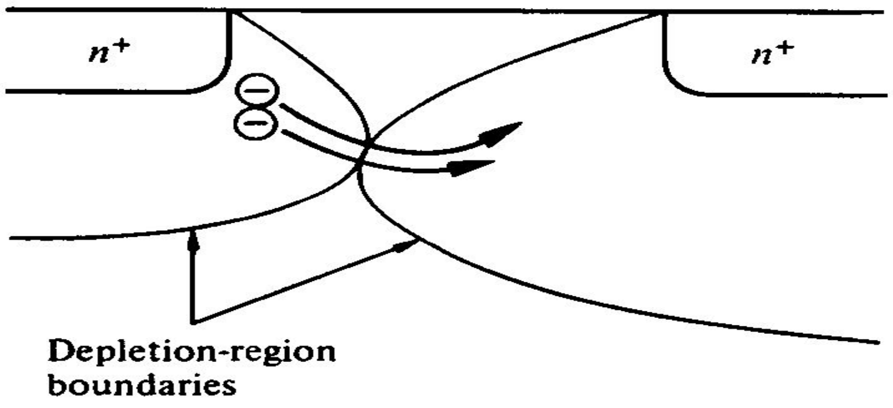
Solution: Increase average channel doping and reduce gate oxide thickness

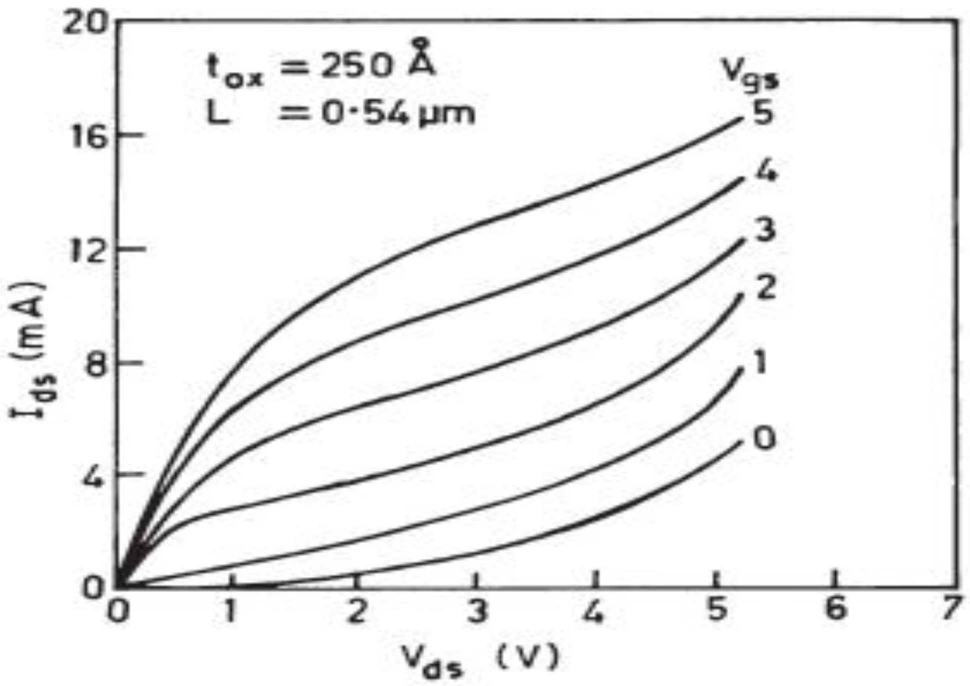
Punch-through

- A large drain bias can cause the drain-junction depletion region to merge with the source-junction depletion region, forming a sub-surface path for current conduction.

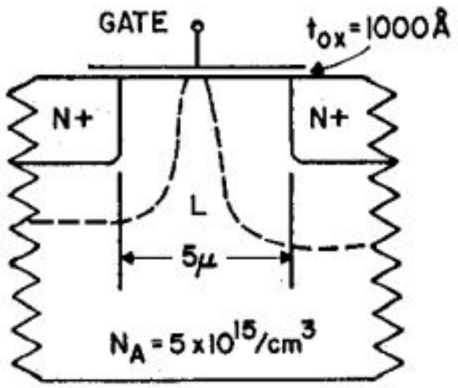
 I_{Dsat} increases rapidly with V_{DS}

- This can be mitigated by doping the semiconductor more heavily in the sub-surface region, *i.e.* using a “anti-punch through” implant.





MOSFET scaling (Dennard 1972)



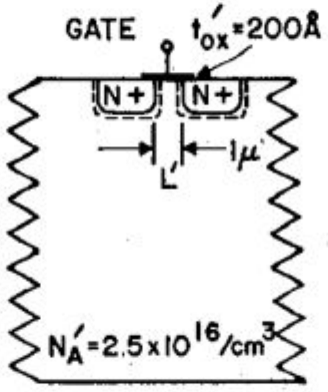
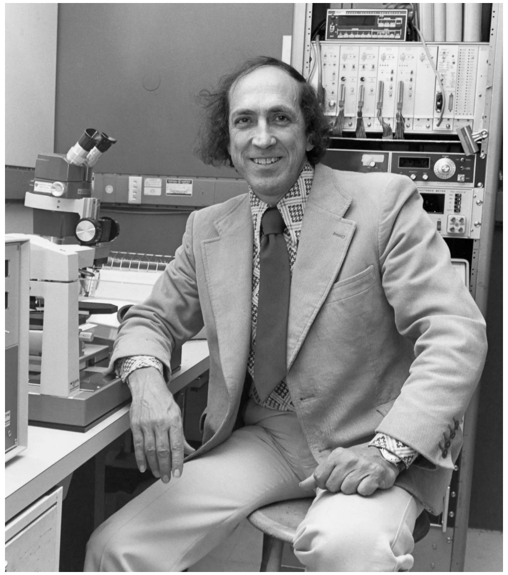


TABLE I

SCALING RESULTS FOR CIRCUIT PERFORMANCE

Device or Circuit Parameter	Scaling Factor
Device dimension t_{ox}, L, W	$1/\kappa$
Doping concentration N_a	κ
Voltage V	$1/\kappa$
Current I	$1/\kappa$
Capacitance $\epsilon A/t$	$1/\kappa$
Delay time/circuit VC/I	$1/\kappa$
Power dissipation/circuit VI	$1/\kappa^2$
Power density VI/A	1



The phenomenon of 'yes, there is free lunch' comes ever so rarely in engineering. The ability to build more transistors in a chip without having to worry about cooling accelerated the availability of compute power for many decades, and is solely responsible for the rapid development of chip technology in the past century.

An added benefit comes from the drop in delay time by a factor of 1/k which means that not only were we able to cram in more transistors with no thermal penalty, but those transistors switched faster too, enabling higher clock speeds. This really is an instance of 'having your cake and eating it too.'

https://www.viksnewsletter.com/p/how-dennard-scaling-actually-works
https://www.nytimes.com/2024/05/16/technology/robert-dennard-dead.html

Scaling Rule of Thumb

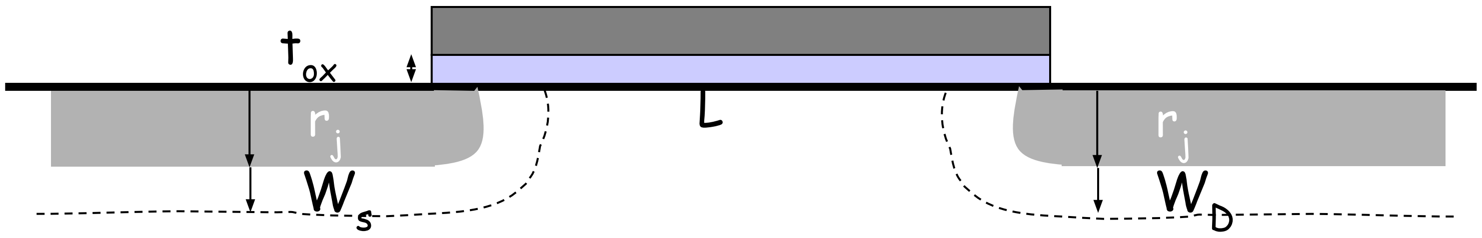
- L_{min} minimum gate length for "long channel" behavior (Empirical relation based on simulation)

$$L_{\min} = 0.4\left[r_j t_{ox}(W_s + W_D)^2\right]^{1/3}$$

  t_{ox} gate oxide thickness in Å

  L_{min}, W_s, W_D, r_j(S/D junction depth) in microns

- A FET is whether electrically long or short really depends upon the process conditions rather than strictly only on the gate length.
- S/D junctions, depletion width (doping), oxide thickness must scale with minimum gate length



Shallow Source and Drain (S/D)

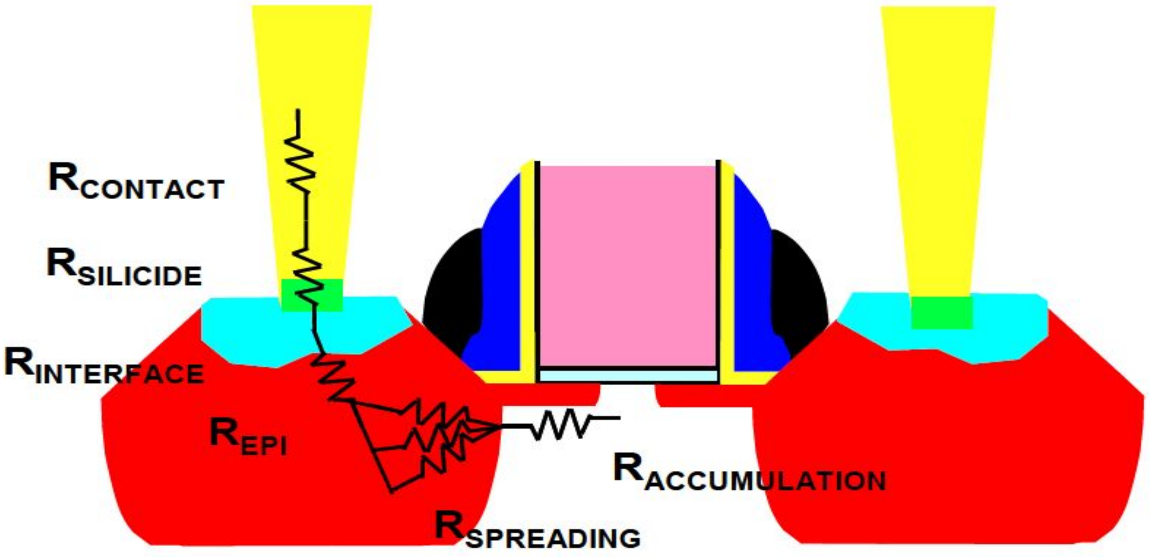
- To minimize the short channel effect and DIBL, we want shallow (small r_j) S/D regions – but the parasitic resistance of these regions increases when r_j is reduced.

$$R_{source}, R_{drain} \propto \rho / W r_j$$

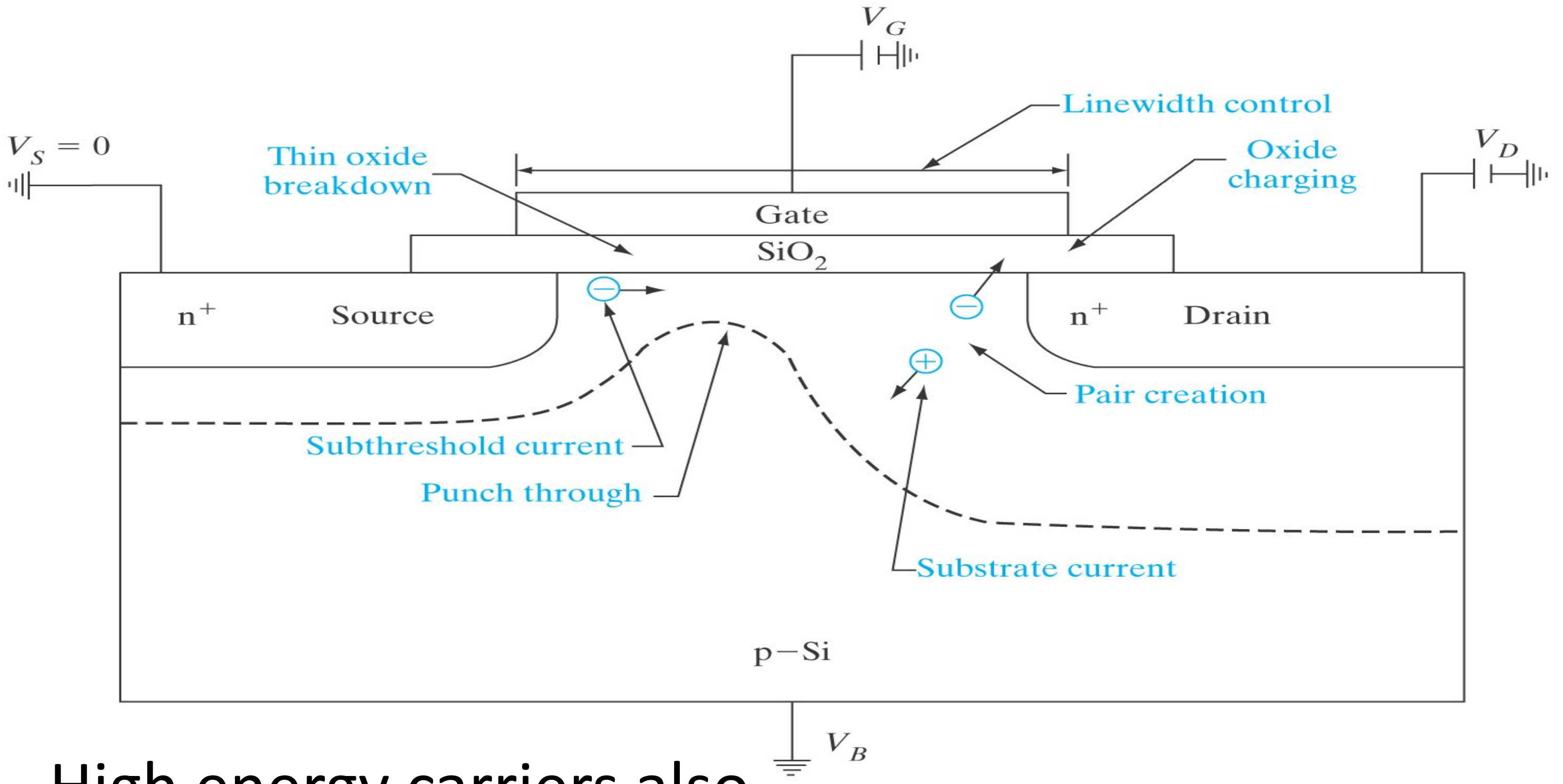
where r = resistivity of the S/D regions

- These regions need to be heavily doped (and better activated) to make the increase in parasitic resistance smal

- One still needs heavily doped deep source/drain to make contacts and reduce the series resistance



Hot carrier effects



High electric field near drain junction increases carrier energy and some of them can get trapped inside oxide and/or create new traps.

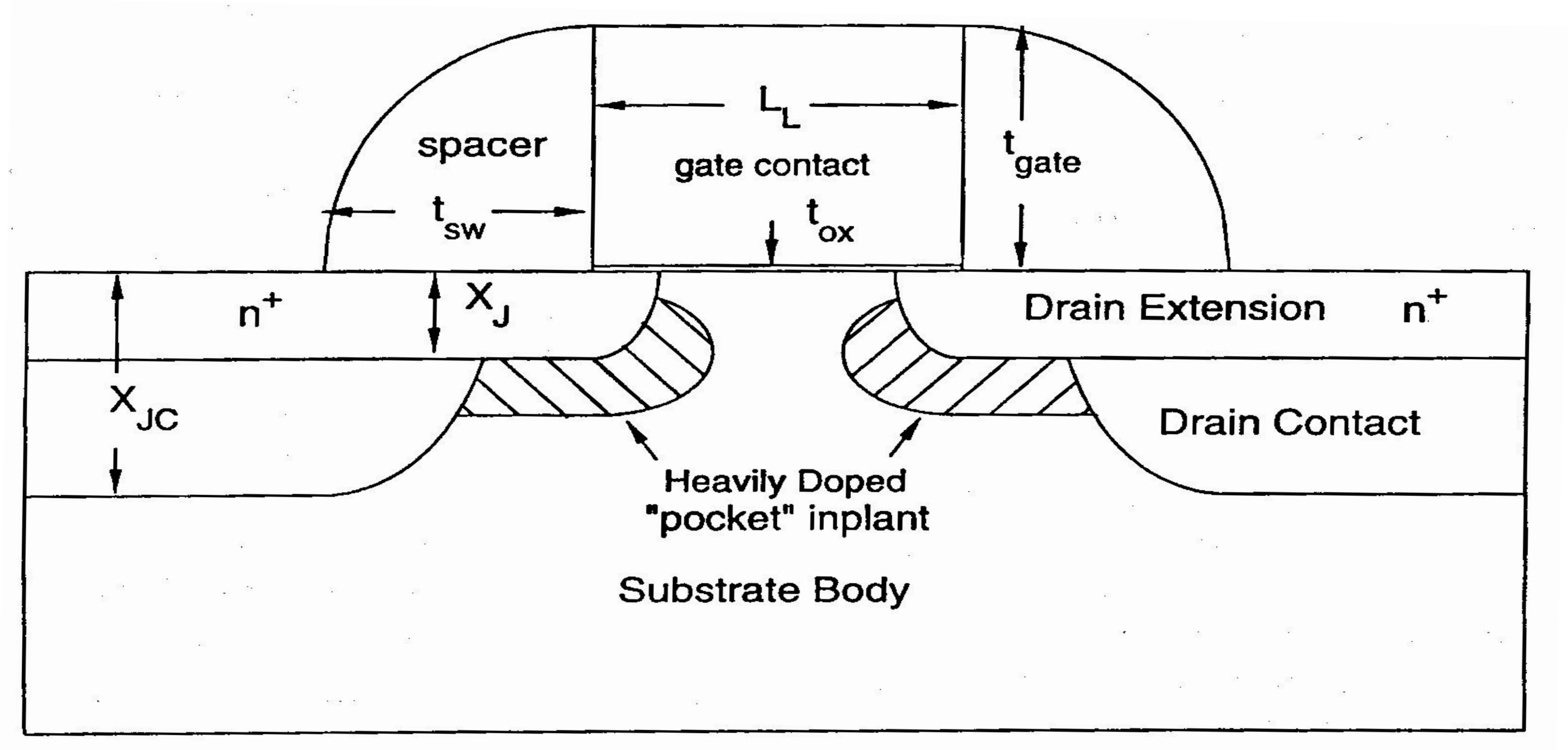
Increases V_T, reduces I_d affecting long term reliability

High energy carriers also create EHPs through impact ionization. The generated holes flow to substrate (I_{sub})

Feature Size (μm)	Power-Supply Voltage (V)	Gate Oxide Thickness (Å)	Oxide Field (MV/cm)
2	5	350	1.4
1.2	5	250	2.0
0.8	5	180	2.8
0.5	3.3	120	2.8
0.35	3.3	100	3.3
0.25	2.5	70	3.6

Plain Vanilla MOSFET structure (before 2003)

- Halo/pocket implants to reduce encroachment of drain depletion region and hence DIBL.
- It is created using tilted implantation along with S/D extension implants
- Spacer defines where to start the deep source/drain



Sub-Threshold Current

- For $|V_G| < |V_T|$, MOSFET current flow is limited by carrier diffusion into the channel region.
- The electric potential in the channel region varies linearly with V_G, according to the capacitive voltage divider formula:

$$\Delta V_C = \frac{C_{oxe}}{C_{oxe} + C_{dep}} \Delta V_G = \frac{1}{m} \Delta V_G$$

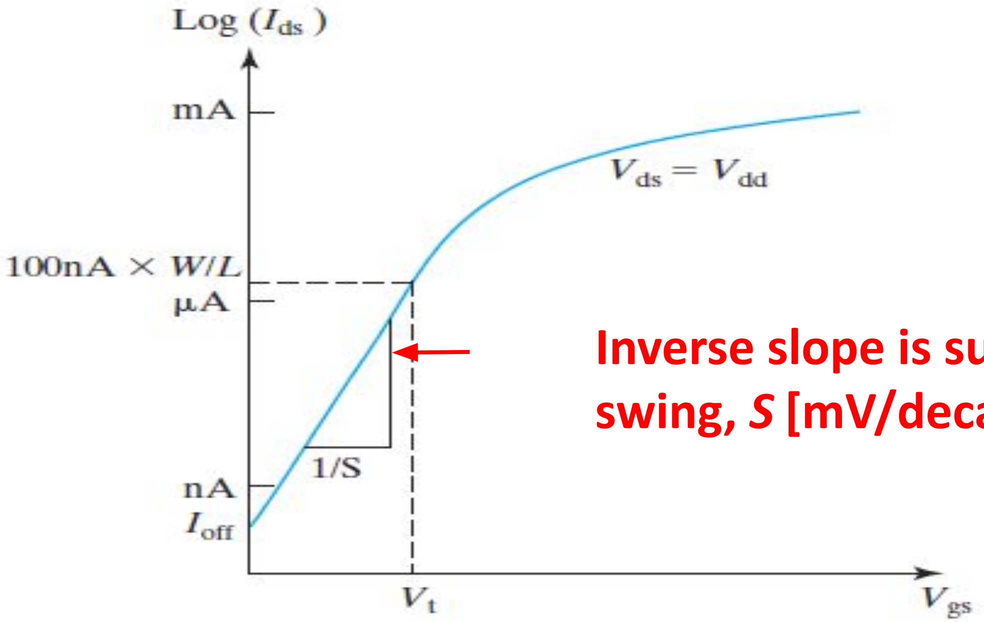
- As the potential barrier to diffusion increases linearly with decreasing V_G, the diffusion current decreases exponentially:

$$I_{DS} \propto e^{qV_G / mkT}$$

Sub-Threshold Swing, *S*

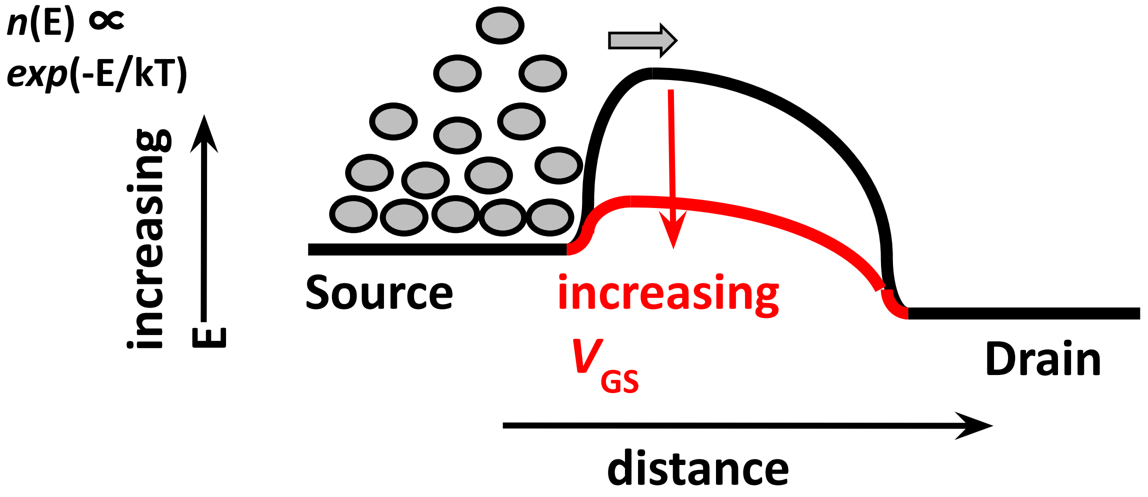
$$S \equiv \left(\frac{d(\log_{10} I_{DS})}{dV_{GS}} \right)^{-1}$$

$$= \frac{kT}{q} ln(10)(1 + \frac{C_{dep,\min}}{C_{oxe}})$$



Inverse slope is subthreshold swing, *S* [mV/decade]

NMOS Energy Band Profile

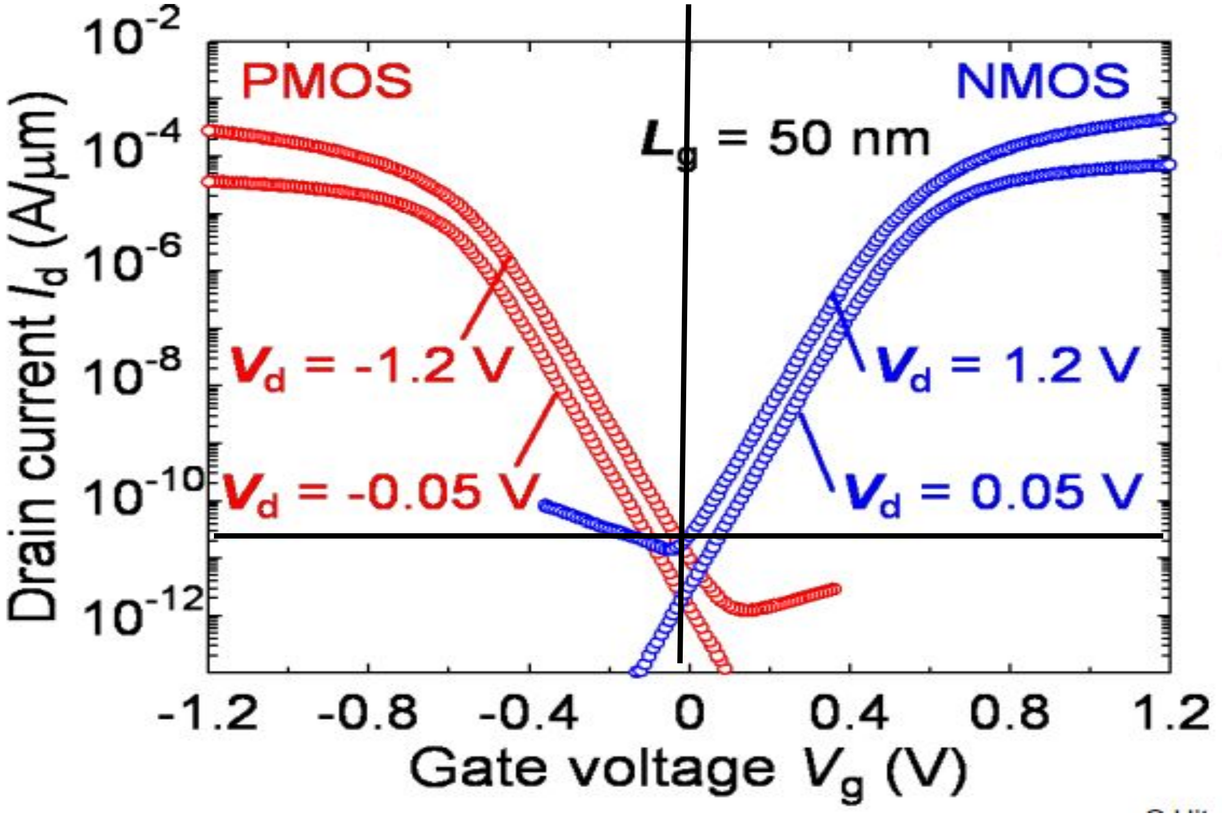


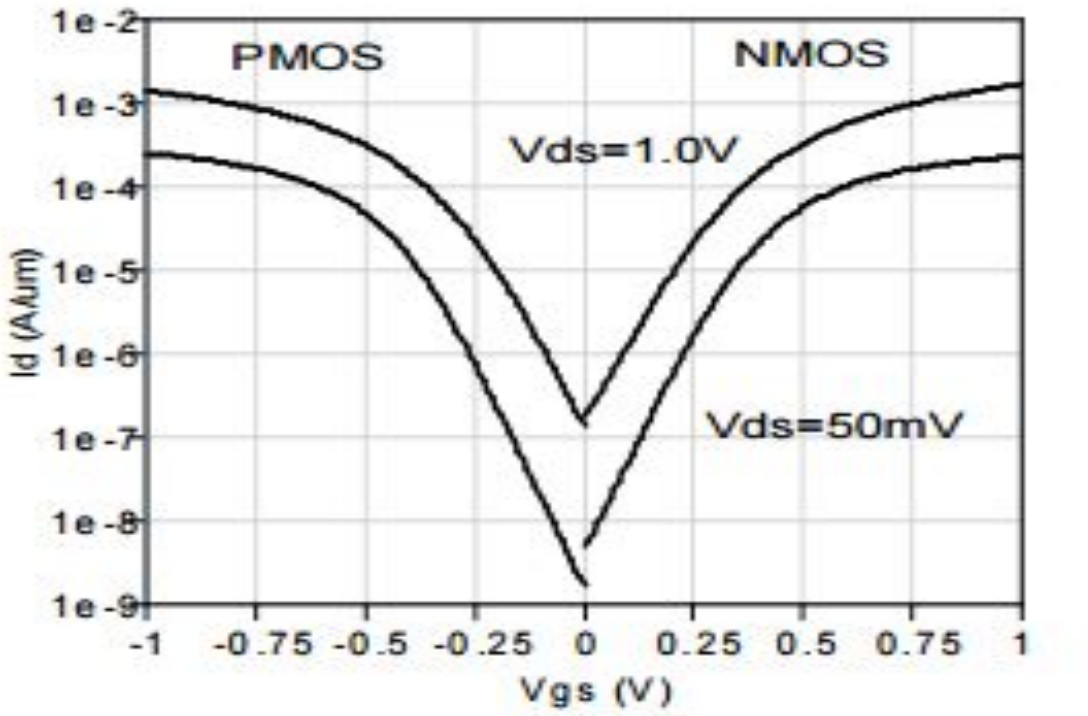
Typical values of S for planar MOSFET – 80-90 mV/decade

Theoretical min at 300 K - ~60 mV/decade

How to reduce *S* ?

Id-Vg (linear and saturation mode)

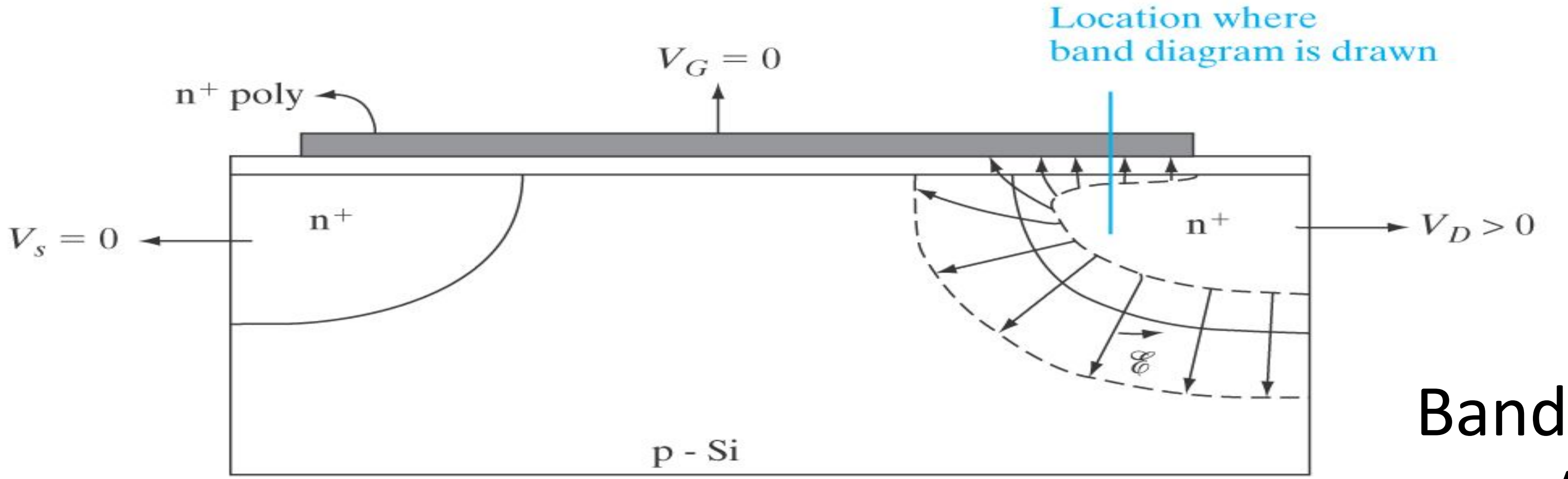


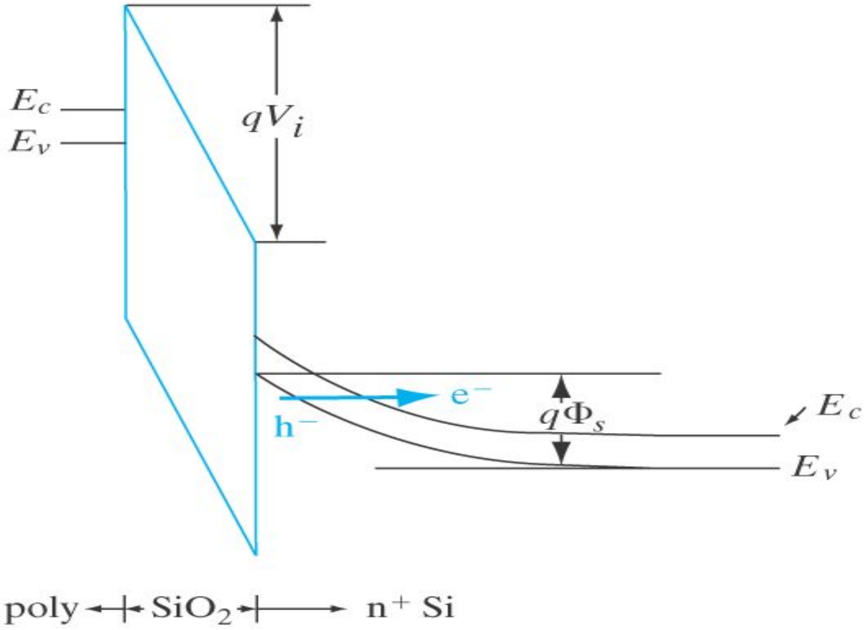


I_{on} (N/P) = 500/300 μA/μm
I_{off}=20 pA/μm

I_{on} (N/P) = 1620/1370 μA/μm
I_{off}=100 nA/μm

Gate Induced Drain Leakage (GIDL)

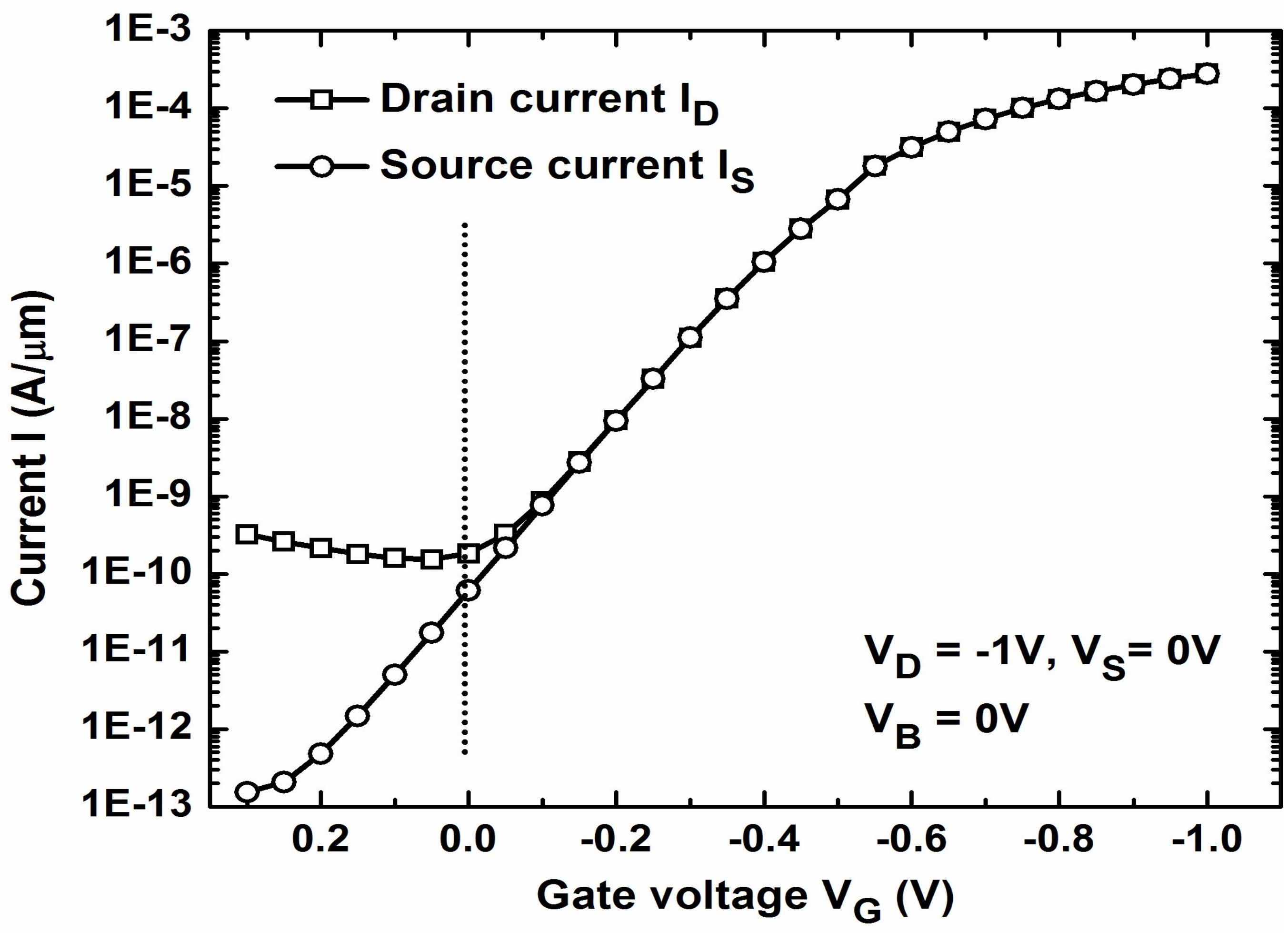




Band to band tunneling in gate/drain overlap region when V_{GD} is –ve

This limits the off current of a FET with high V_T. In such devices, off current (< 100pA/μm) could be dominated by GIDL compared to sub threshold leakage current

GIDL in a PFET



GIDL makes $I_S \neq I_D$

One mayn't see this in a FET with low $V_{T,}$ which has I_{off} > 10 nA/μm

V. Tiwari et al, *IEEE Trans. Electron Devices*, vol. 61, no. 5, pp.1270-1277, May 2014

Gate Leakage current due to tunneling

IEEE ELECTRON DEVICE LETTERS, VOL. 18, NO. 5, MAY 1997

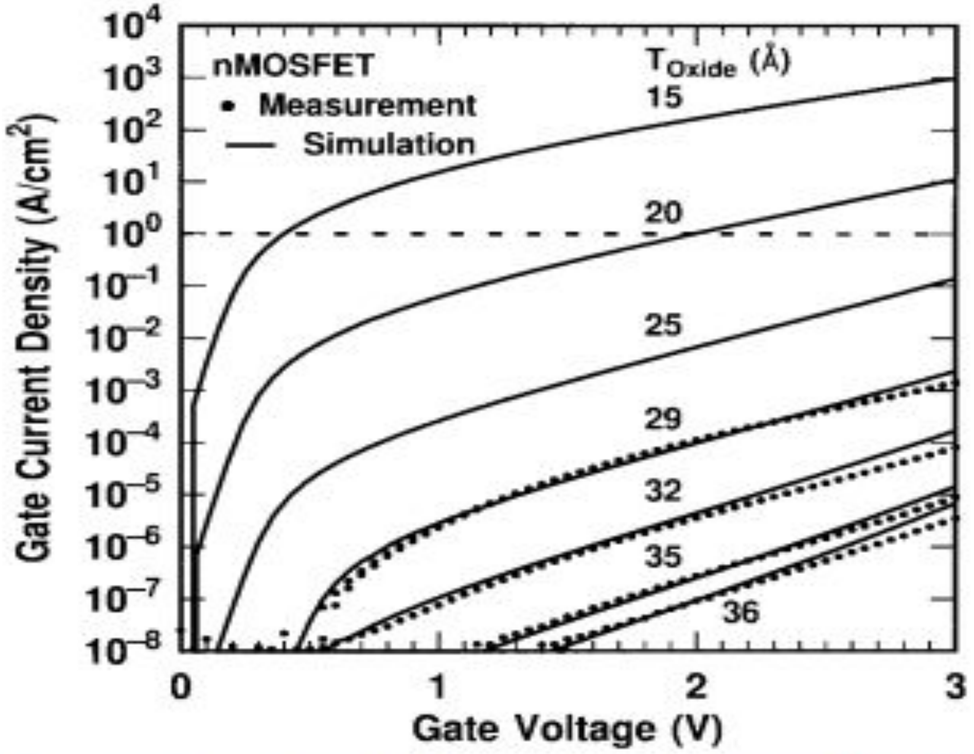
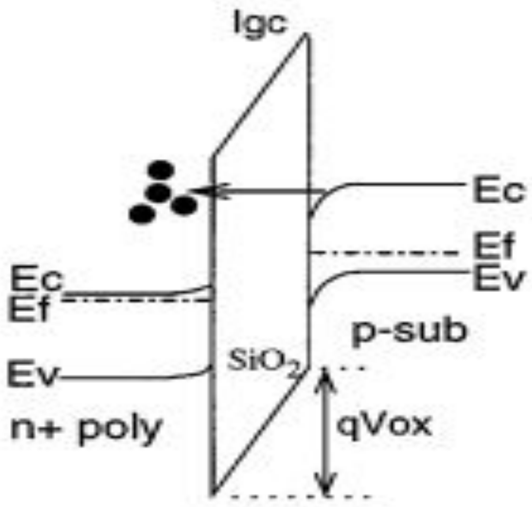
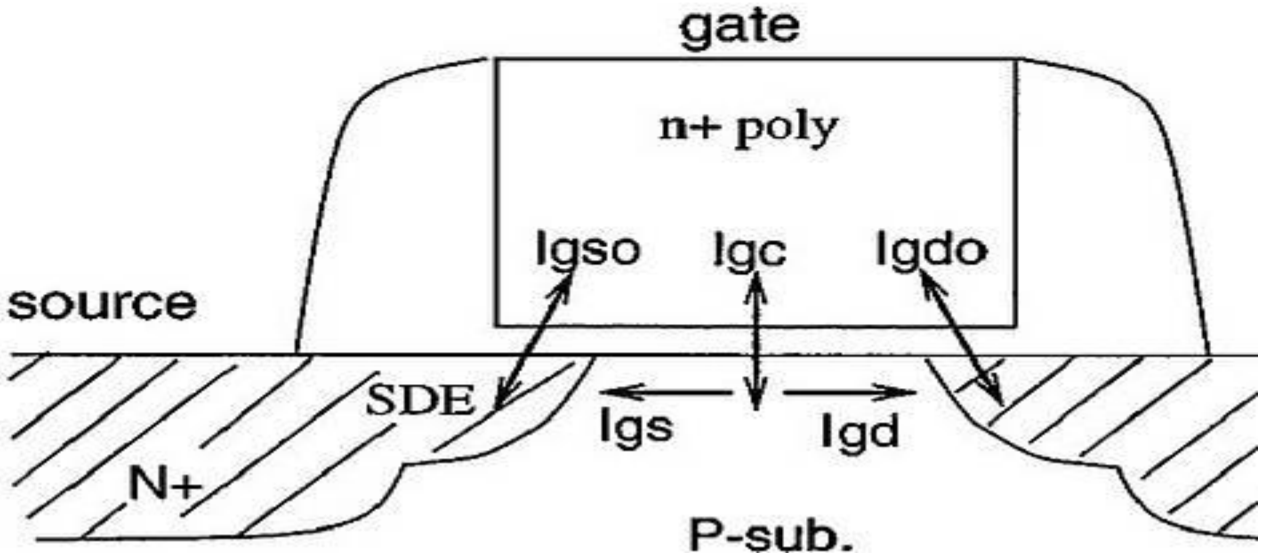


Fig. 2. Measured and simulated I_G–V_G characteristics under inversion conditions of four nMOSFET's. The dotted line indicates the 1 A/cm^2 limit for leakage current as discussed in the text.

For 2Ǻ reduction in gate oxide thickness, current increases by 10X

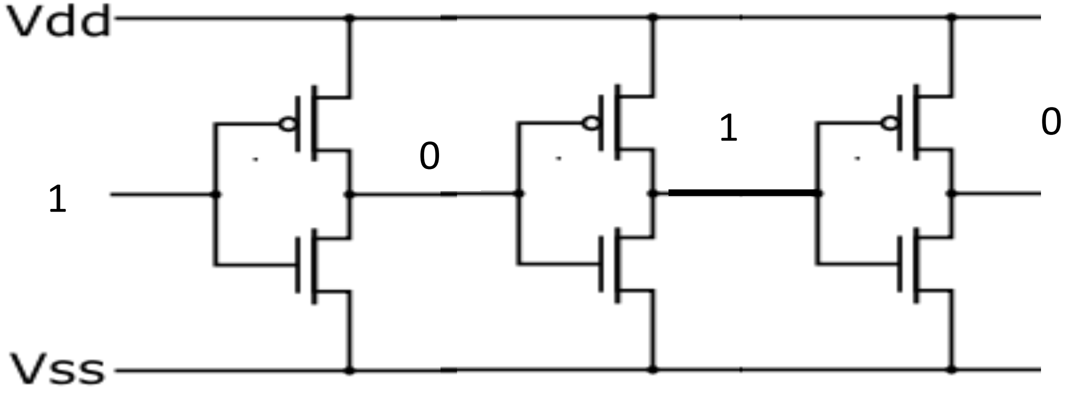


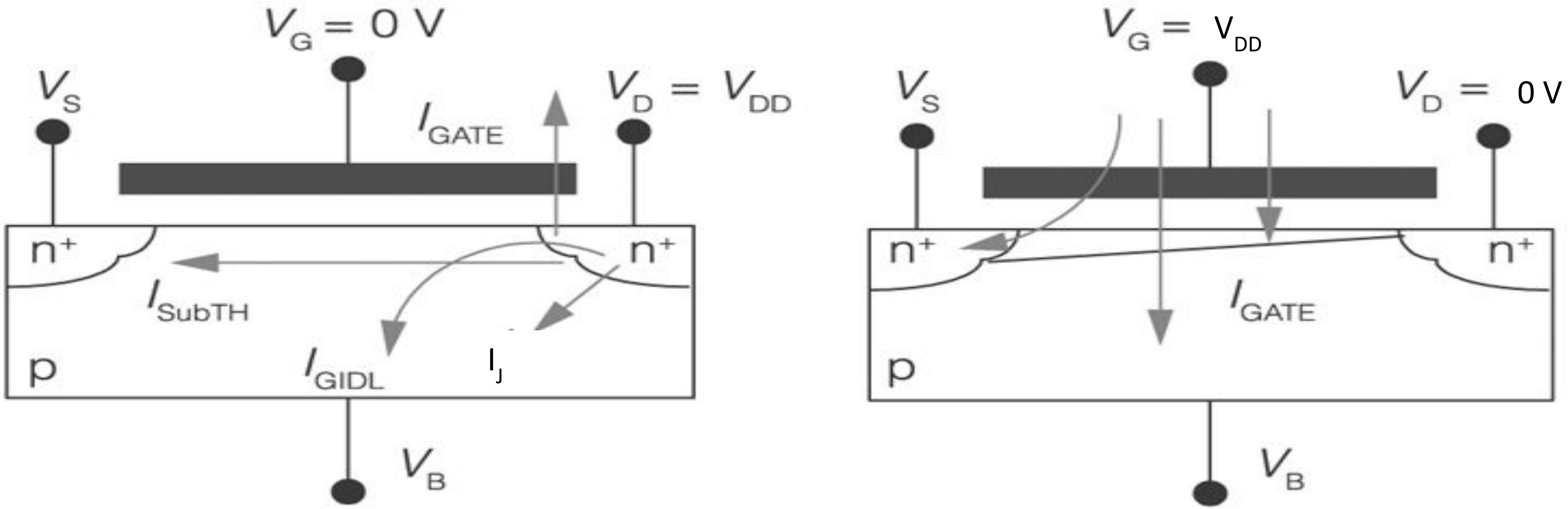


Tunneling a) to/from channel and b) through overlap region

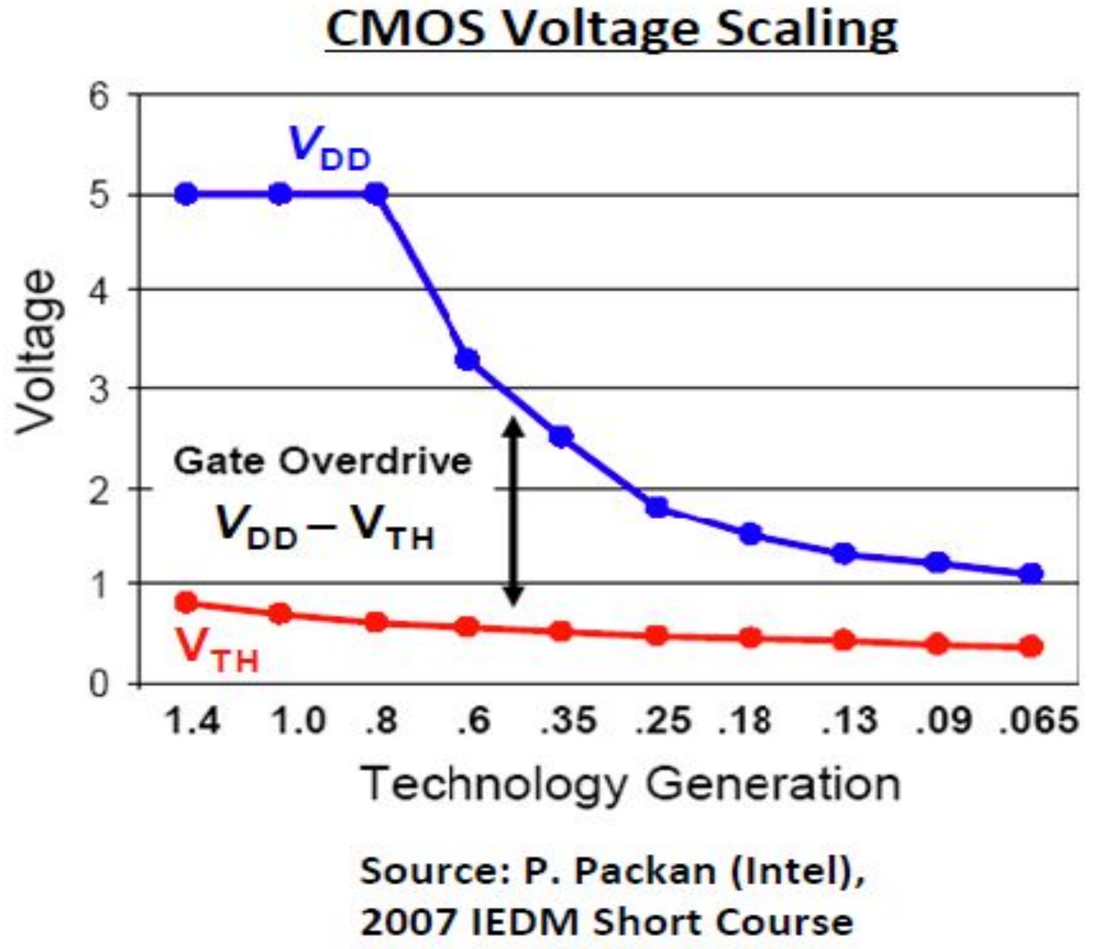
CMOS Inverter

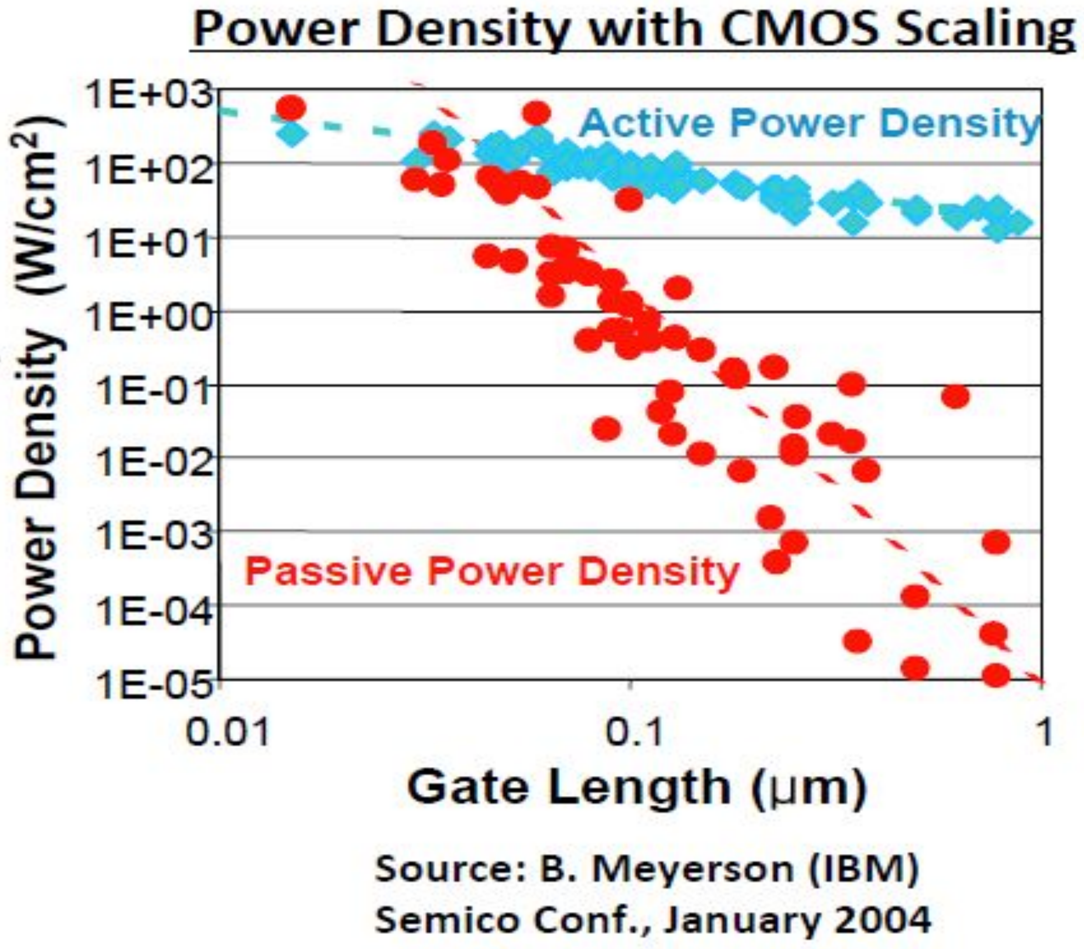
Identify leakage components causing static power dissipation in a CMOS inverter





The power problem in classical scaling





1. Gate length scaling increases source-drain leakage exponentially
☹ Prevents scaling of threshold voltage prevents scaling of supply voltage.
2. Gate dielectric scaling increase gate leakage exponentially
3. Industry diverged into High Performance and Low Power at 90nm
☹ Prevents scaling of T_{ox} degrades channel control increases sub-Vt leakage

End of clock rate wars and rise of multi-core designs

www.eetimes.com/document.asp?doc_id=1144041

IBM flexes Power4 as it ships Regatta servers

IBM flexes Power4 as it ships Regatta servers

David Lammers
10/4/2001 06:07 PM EDT
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AUSTIN, Texas — IBM Corp.'s high-end "Regatta" Unix servers — ranging from an eight-way, 1.1-GHz system, priced at $450,000 up to a 32-way system that hits supercomputer-level speeds — are ready to ship, the company's server division announced Thursday (Oct. 4) .

The server's building block is a palm-sized eight-way module, each holding four Power4 dice. Each Power4 chip includes two 64-bit, 1-GHz processor cores; 64-kbytes of instruction cache; 32-kbytes of data cache; and an L2 cache of 1.4-Mbytes, for a total of 174 million transistors.

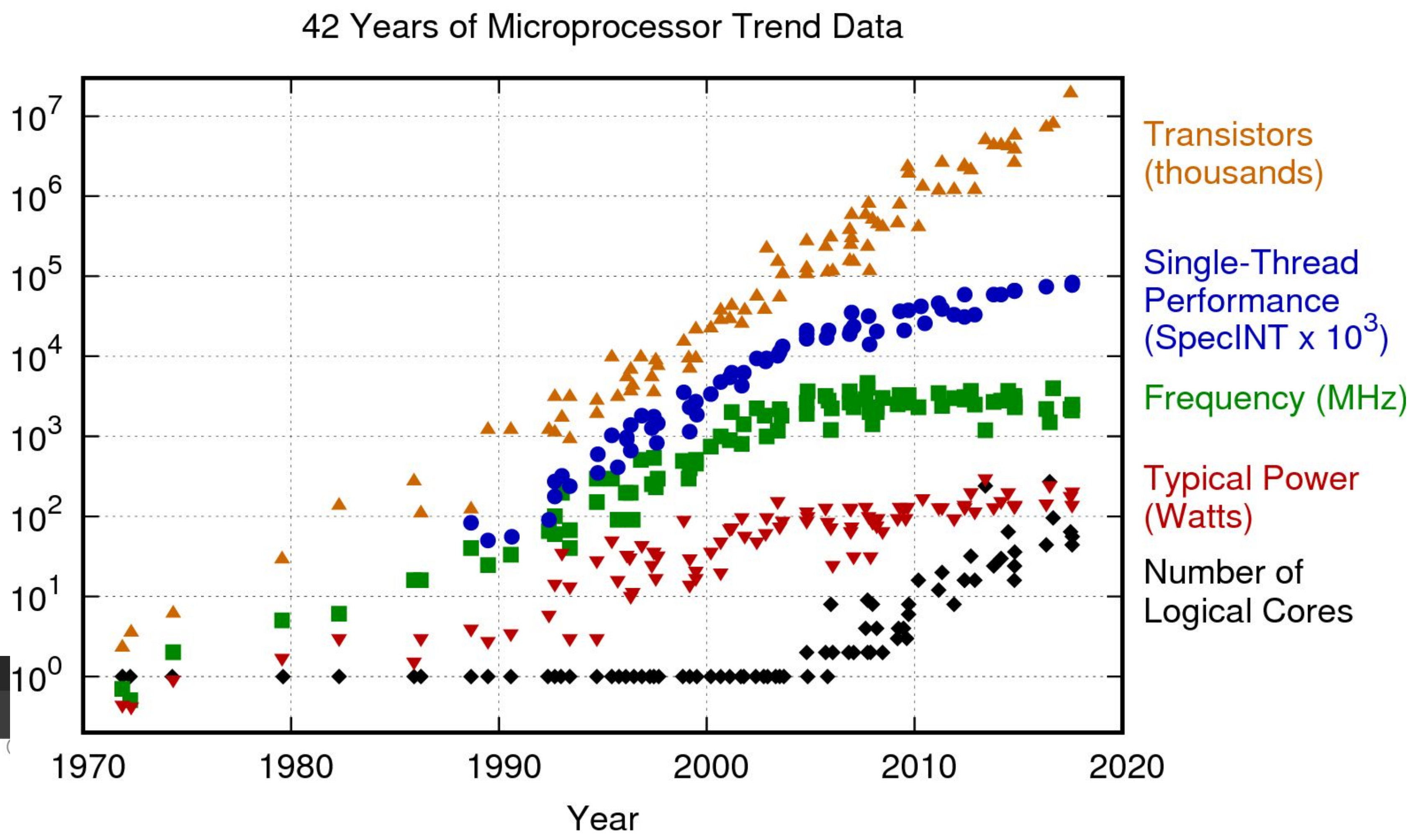
infoworld.com/article/2207567/intel-decides-two-cores-are-better-than-one.html

intenance reque... Adobe Acrobat

by Tom Krazit

Intel decides two cores a[re] better than one

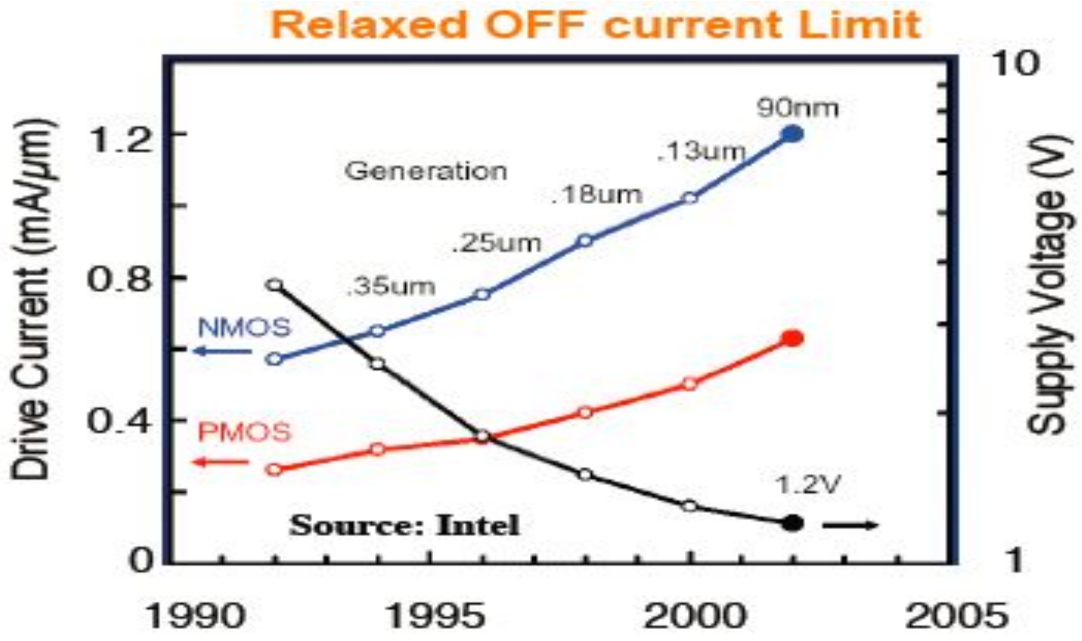
News
May 7, 2004 • 5 mins

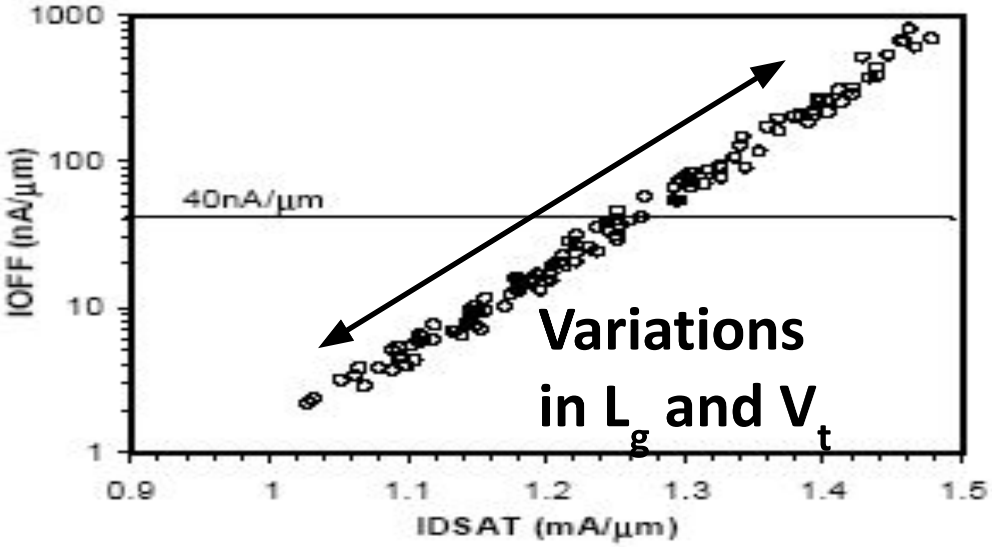


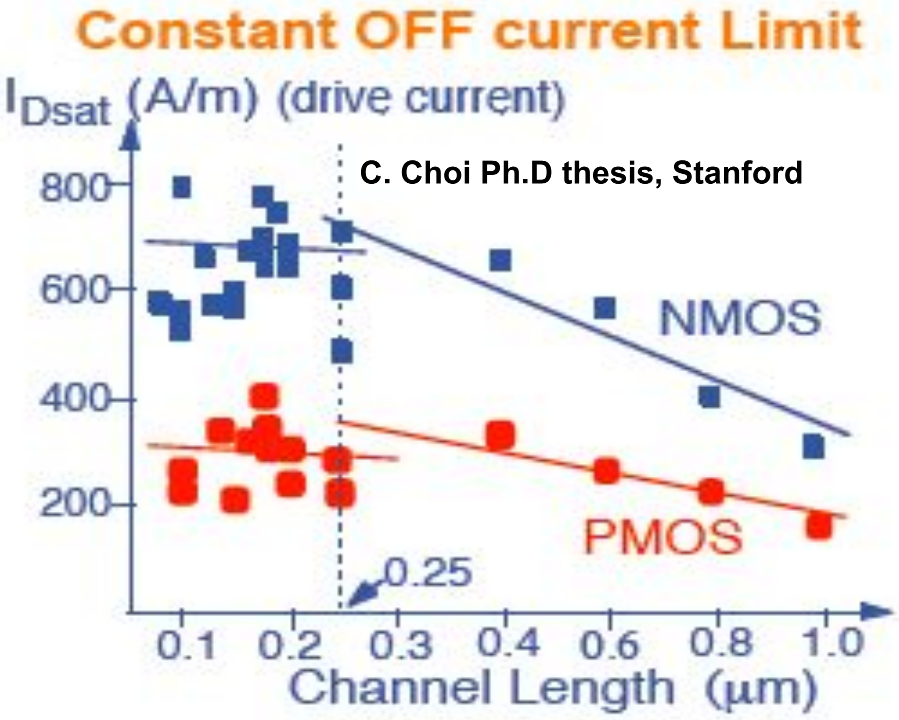
Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2017 by K. Rupp

https://www.karlrupp.net/wp-content/uploads/2018/02/42-years-processor-trend.png

Saturation of I_{Dsat}

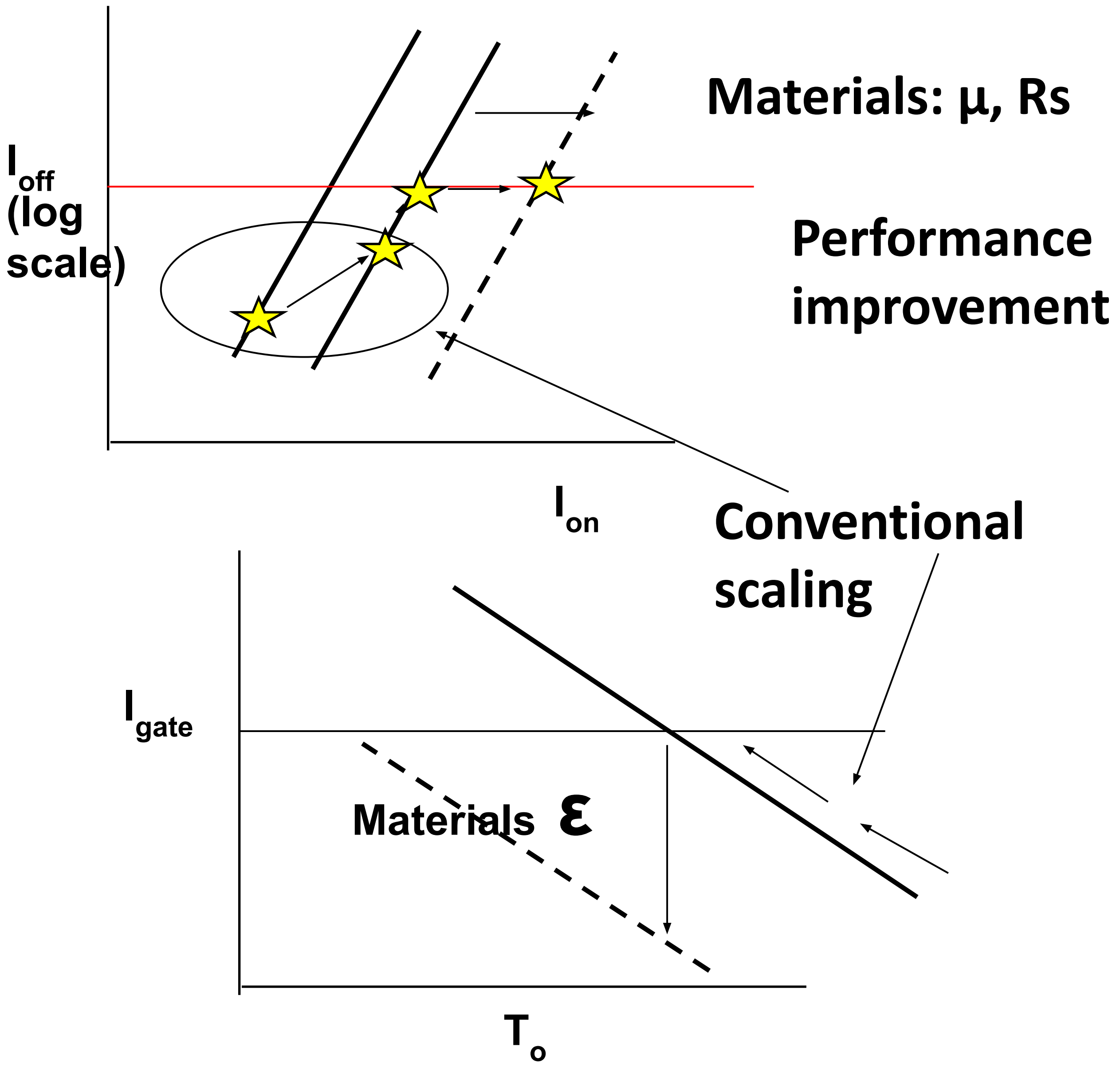






Increase in drive current @ constant I_{off} is needed.

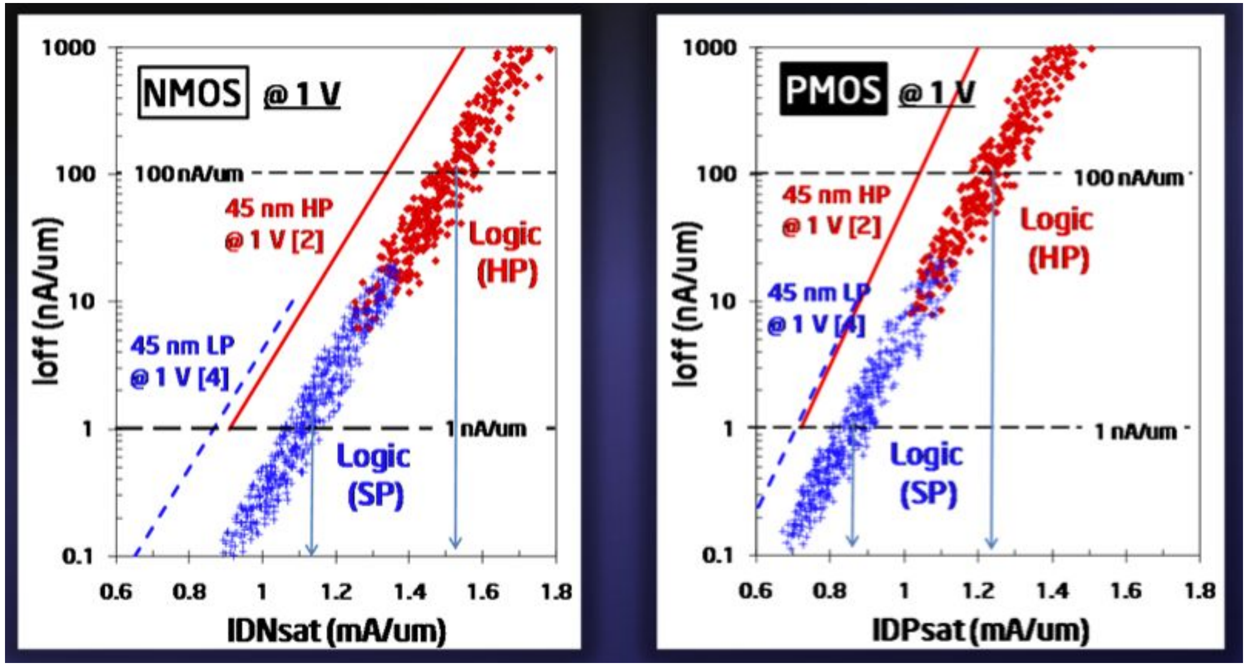
The scaling scenario



Classical scaling stopped at 130nm (circa 2001)

Further scaling is possible by adding new performance enhancing elements

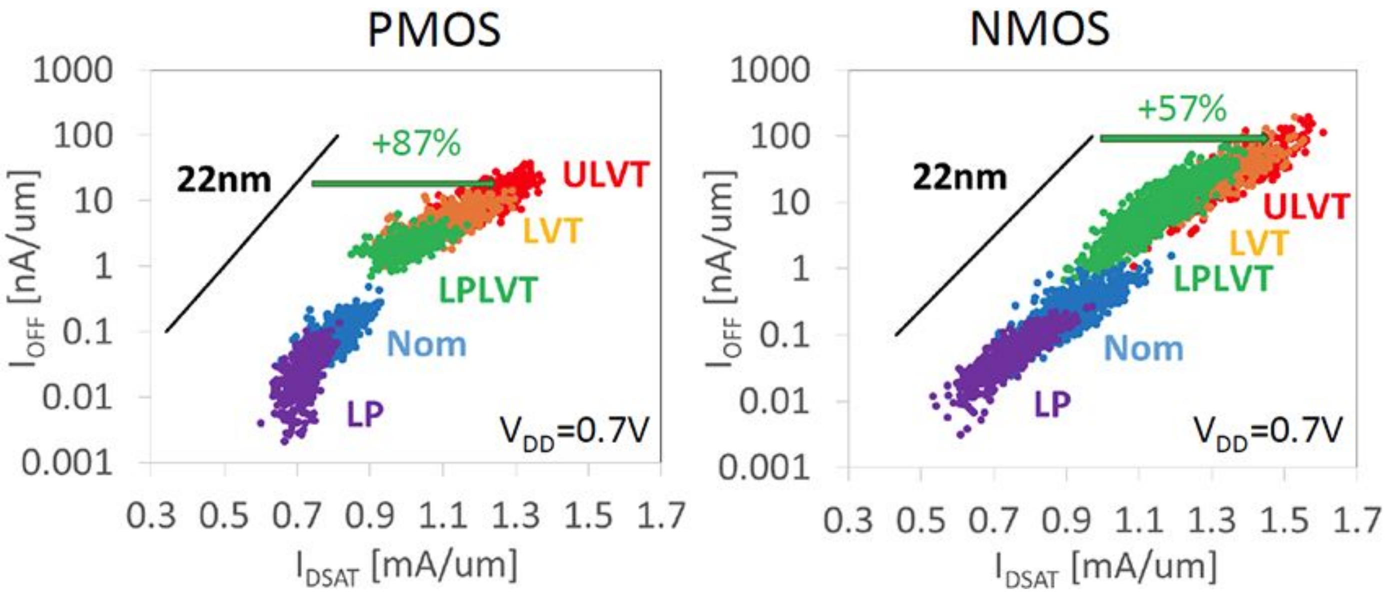
Transistor performance comparison across nodes/nodelet



Intel 45nm-32nm comparison (20-30% improvement) IEDM 2009

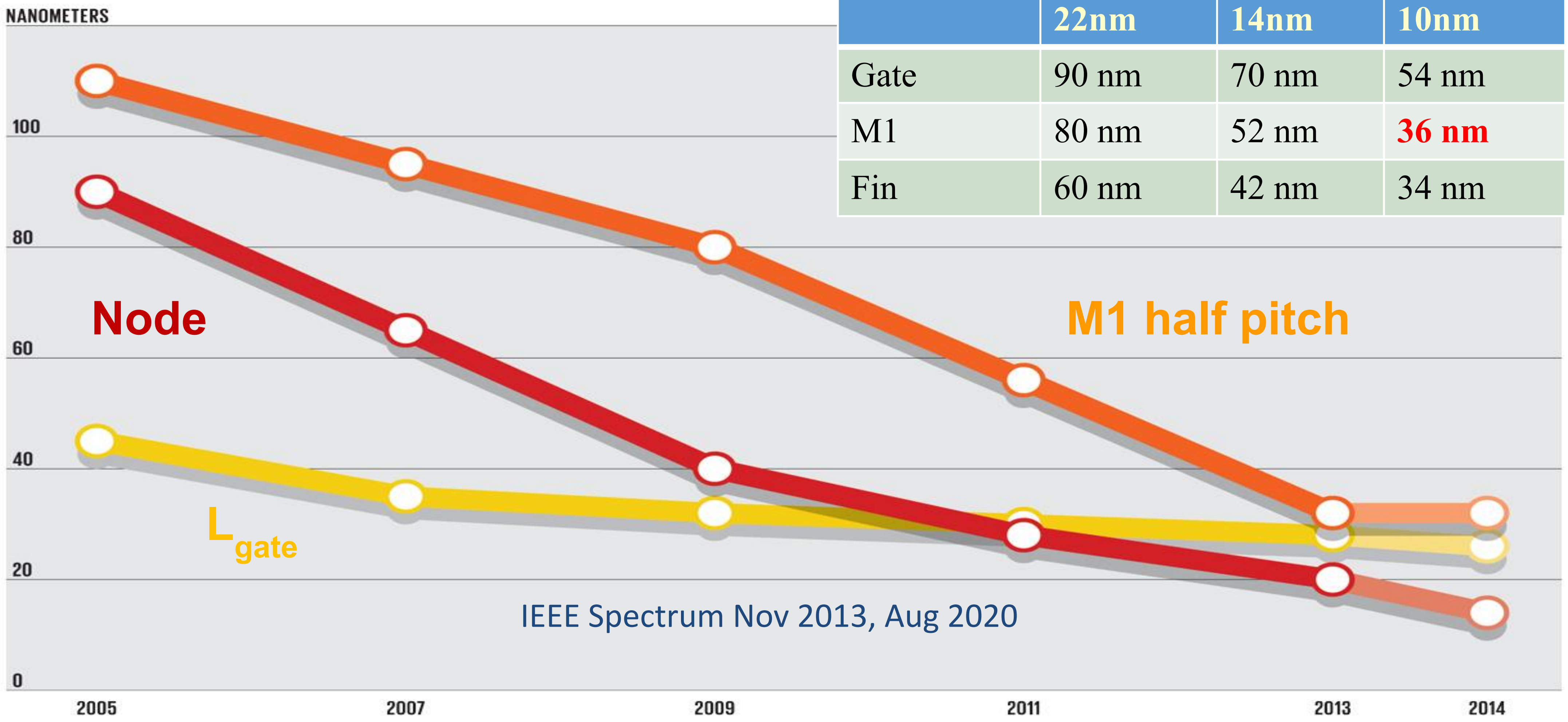
	22nm SoC	14nm SoC	22FFL
Fin Height	34nm	42nm	42nm*
Fin Width	8nm	8nm*	8nm*
Fin Pitch	60nm	42nm	45nm
Gate Pitch	90nm/108nm	70nm/84nm	108nm/144nm
Min. Metal Pitch	80nm	52nm	90nm
Cell Height	840nm	399nm	540nm

Intel 22nm-22FL comparison IEDM 2011/2018



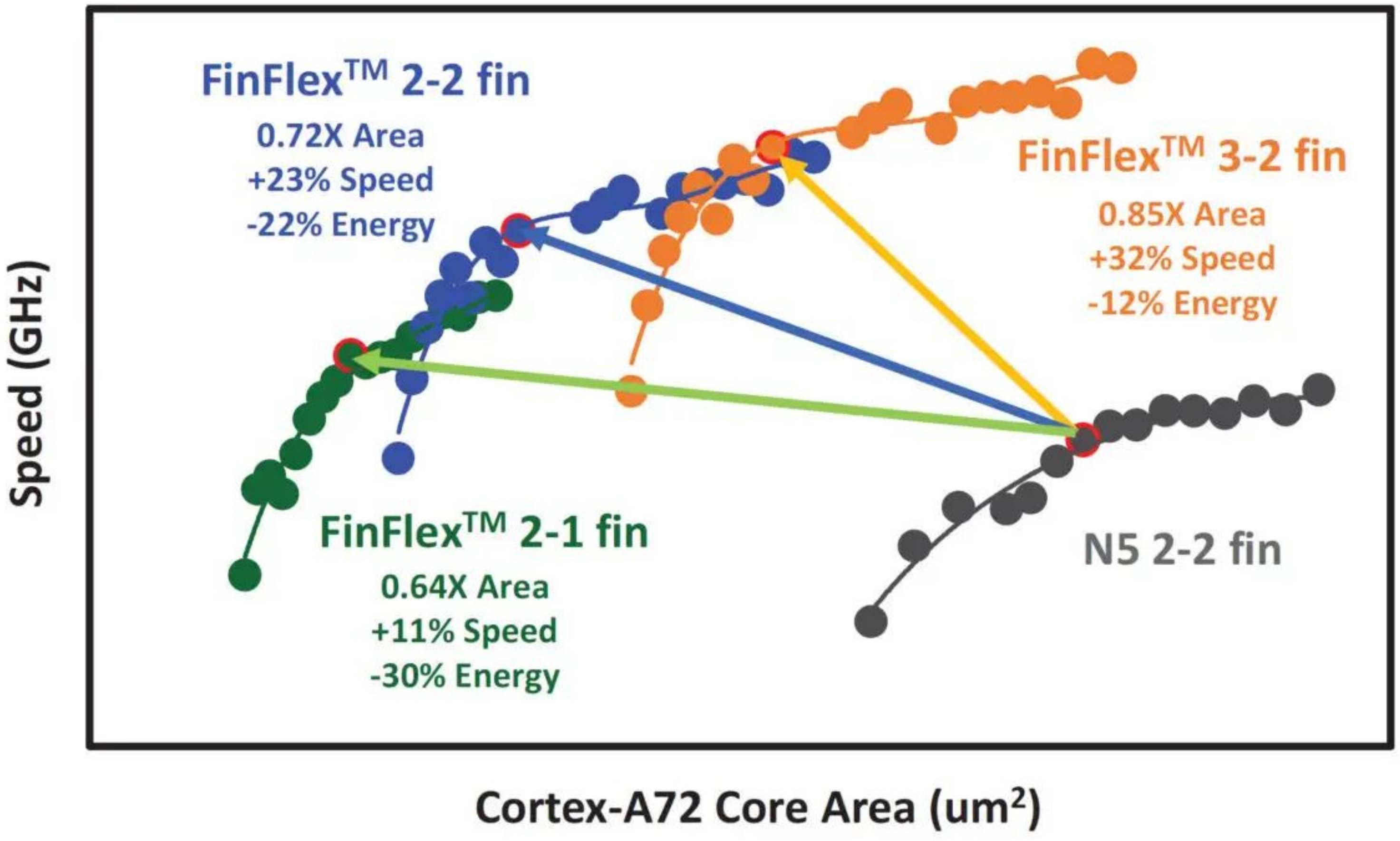
What does node name means ?

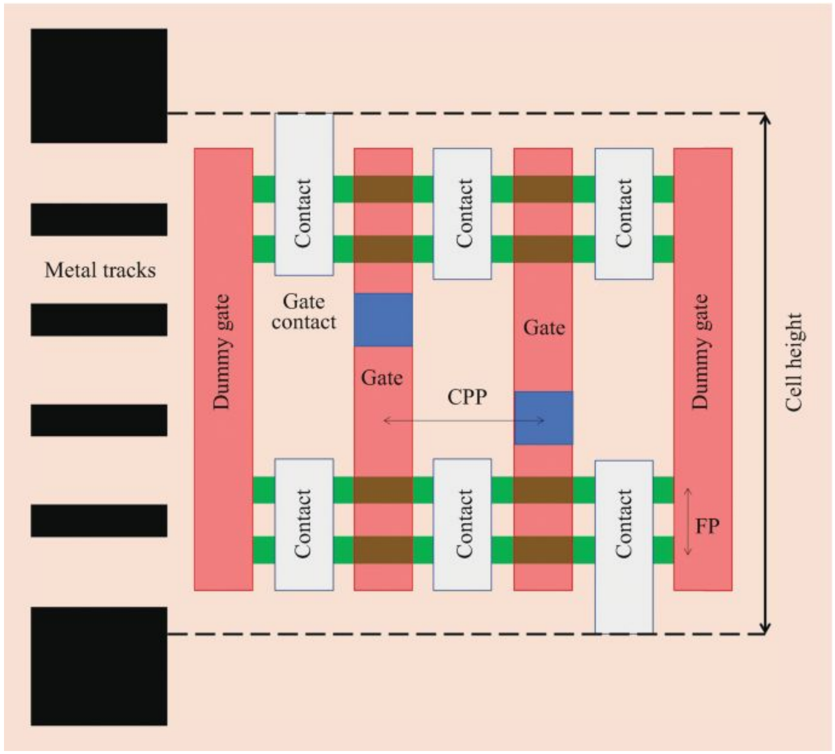
Pitch	Intel 22nm	Intel 14nm	Intel 10nm
Gate	90 nm	70 nm	54 nm
M1	80 nm	52 nm	**36 nm**
Fin	60 nm	42 nm	34 nm



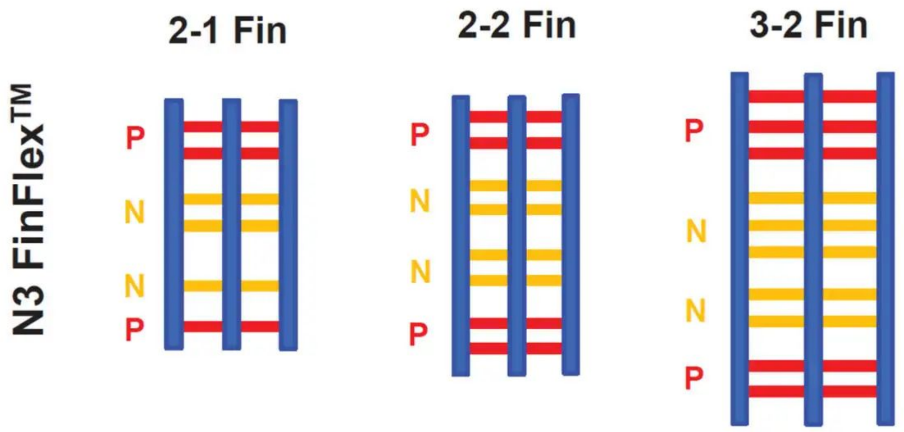
Node name doesn't represent any feature size on the wafer or in the design rules. There are also nodelets or half nodes !!

What does nodelet means ?





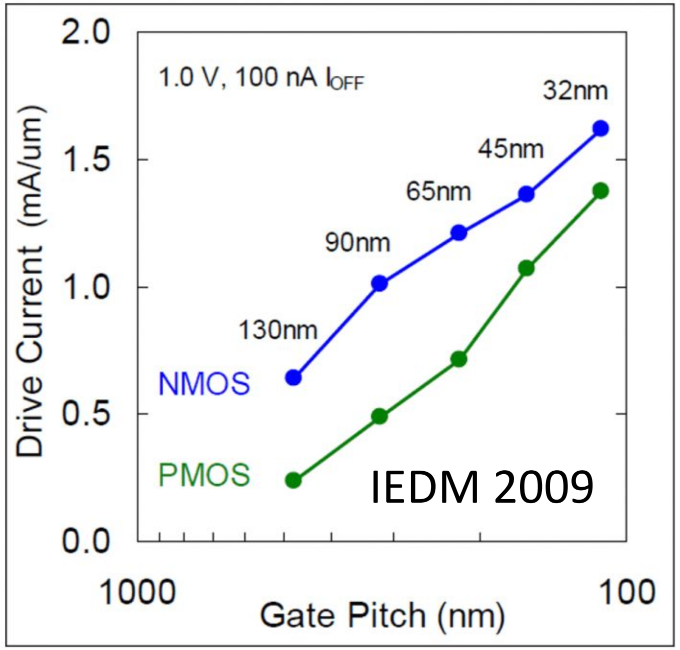
TSMC 3nm IEDM 2022

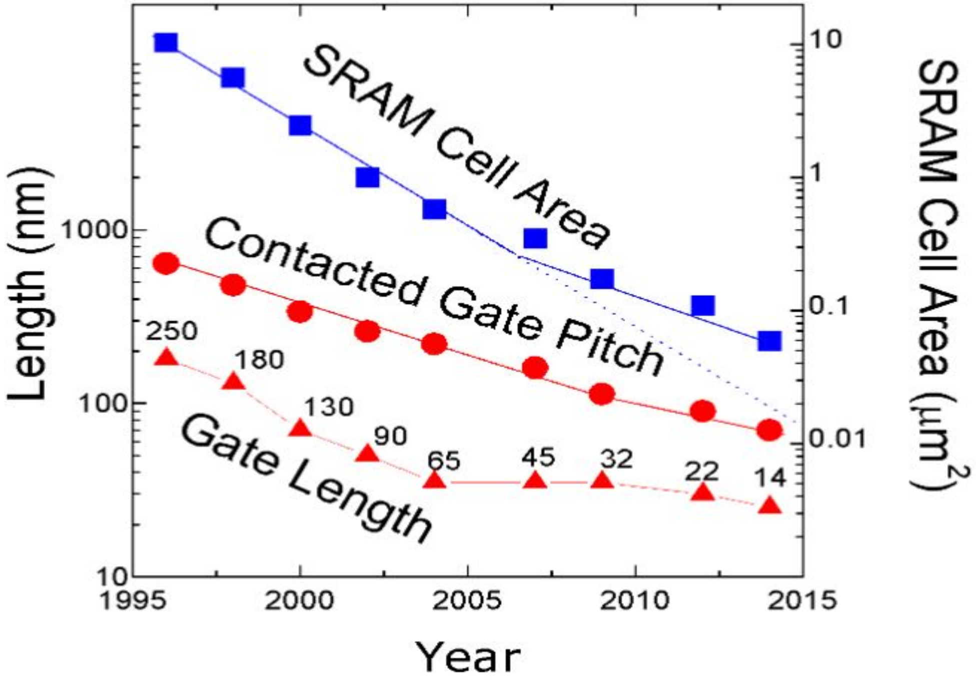


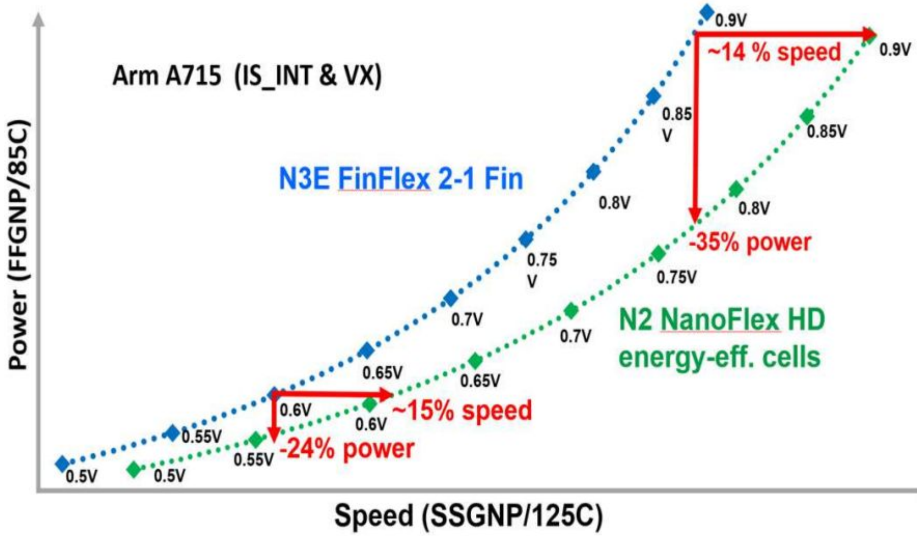
Pitch scaling @ higher FLOPS/Watt

- At a very high level, desirable product features are
  - Performance per power (FLOPS/Watt)
  - More functionality per area at the same cost

- The major factor which enabled this was continuous realization of smaller MOSFETs with larger drive currents



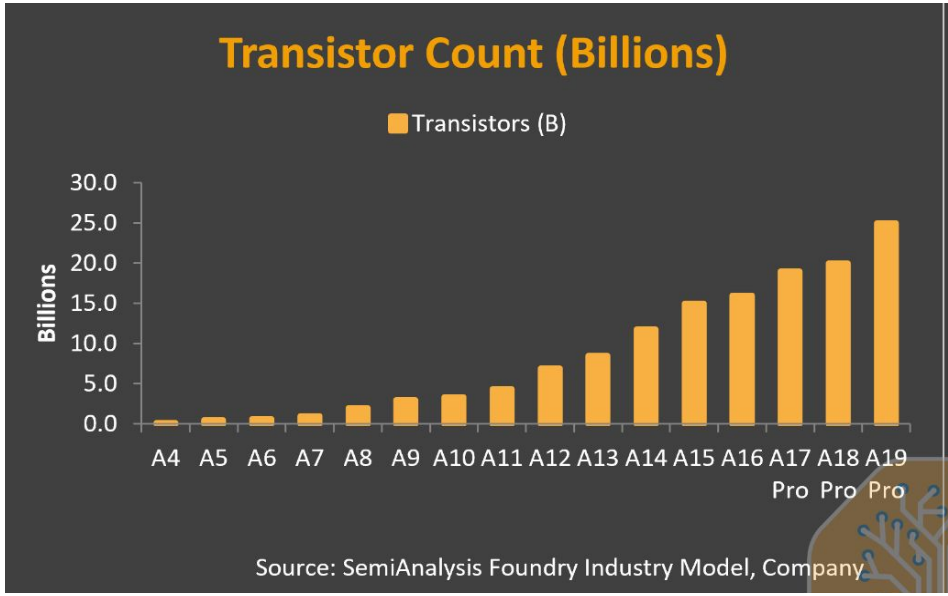


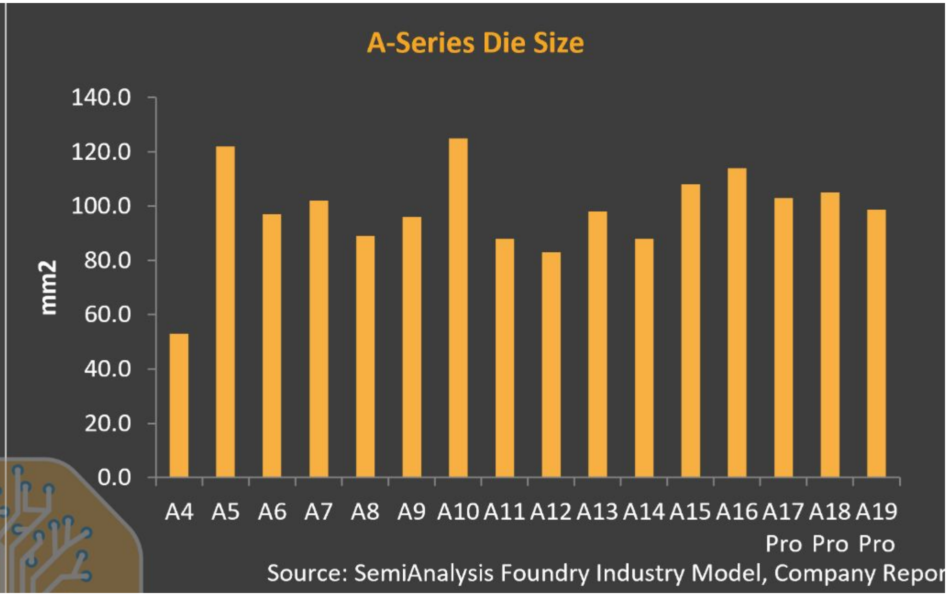


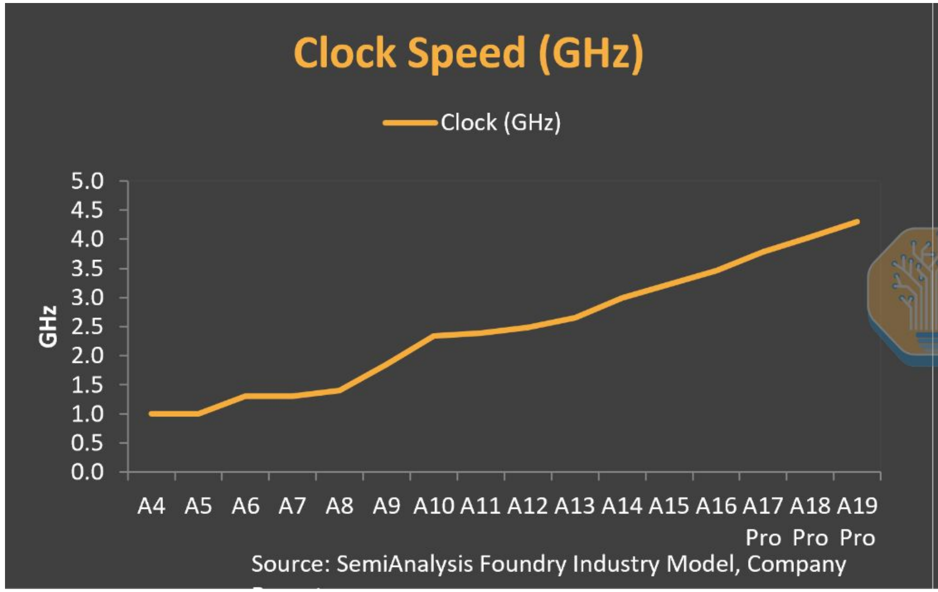
TSMC @ IEDM 2024

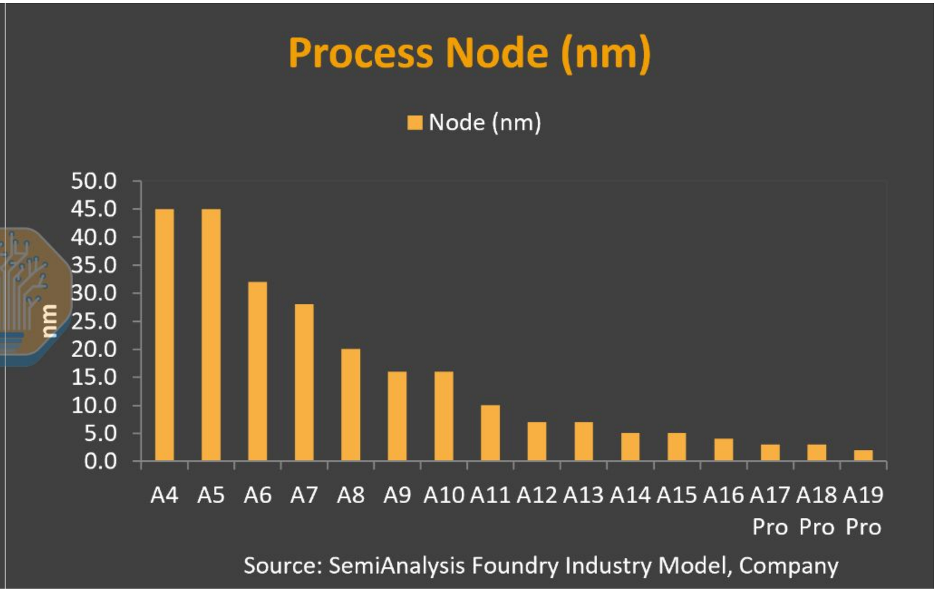
https://www.linkedin.com/pulse/cmos-density-scaling-cppmxp-metric-ali-khakifirooz
https://www.intel.com/content/dam/doc/technology-brief/32nm-logic-high-k-metal-gate-transistors-presentation.

Apple processors



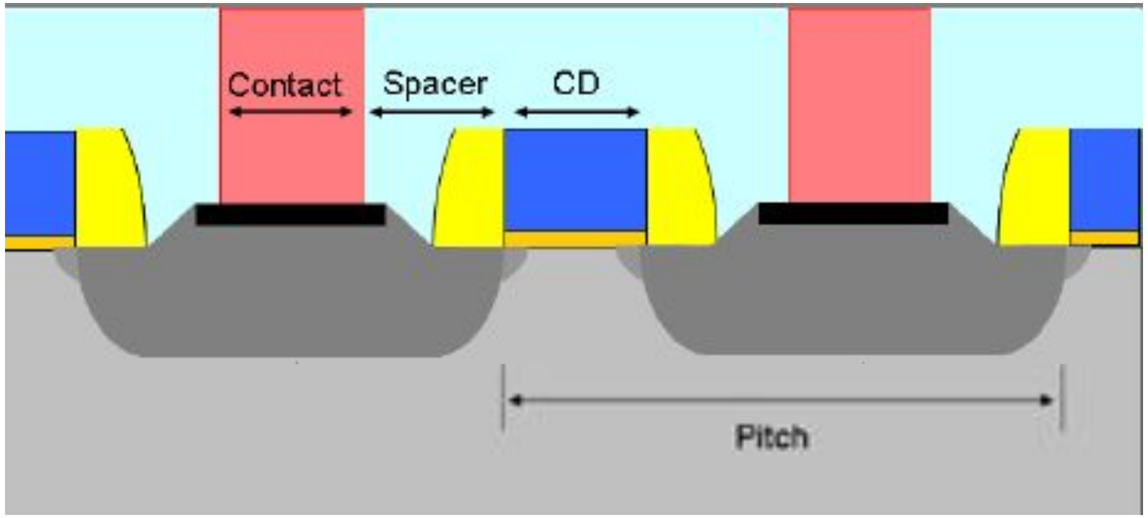






https://newsletter.semianalysis.com/p/apple-tsmc-the-partnership-that-built

Why gate length scaling is still important?



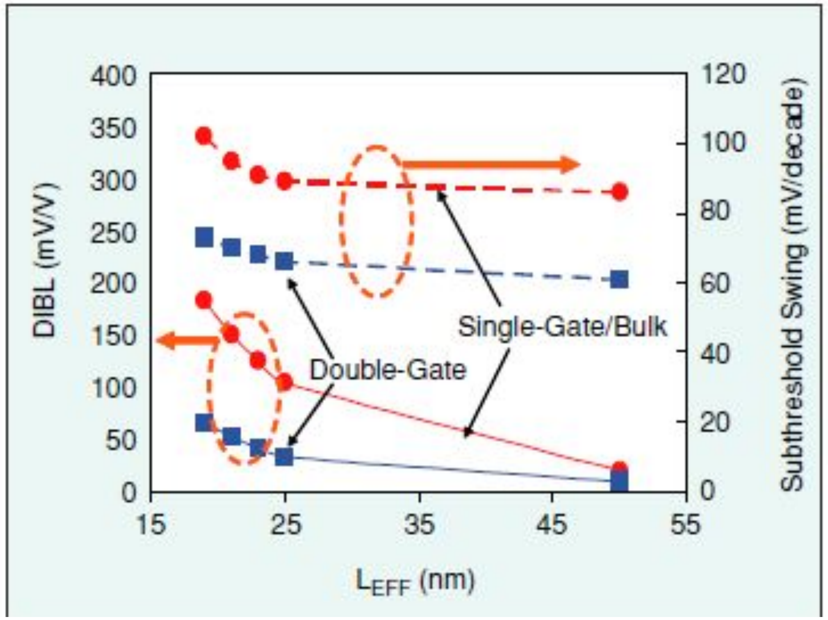
- Contacted Gate pitch=Gate length +2*(Poly-Contact spacing) + Contact width

- Before 90nm node, all three terms in RHS used to scale more or less equally to reduce the gate pitch by ~0.7x every node

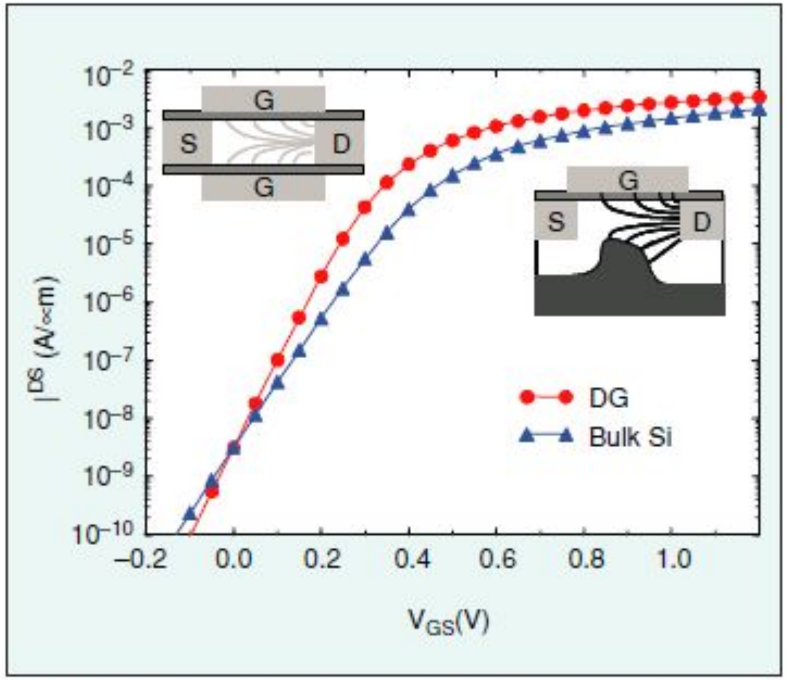
- However, gate length scaling has slowed down after 90nm node resulting contact and poly-contact spacing should scale more to accommodate a longer than desired gate. This results in higher gate & gate-contact capacitance/resistance, poor yield due to poly/contact shorts etc
- Single gate is not able to control the channel below a physical gate length of 30 nm (very large I_{OFF}, DIBL, subthreshold slope etc)

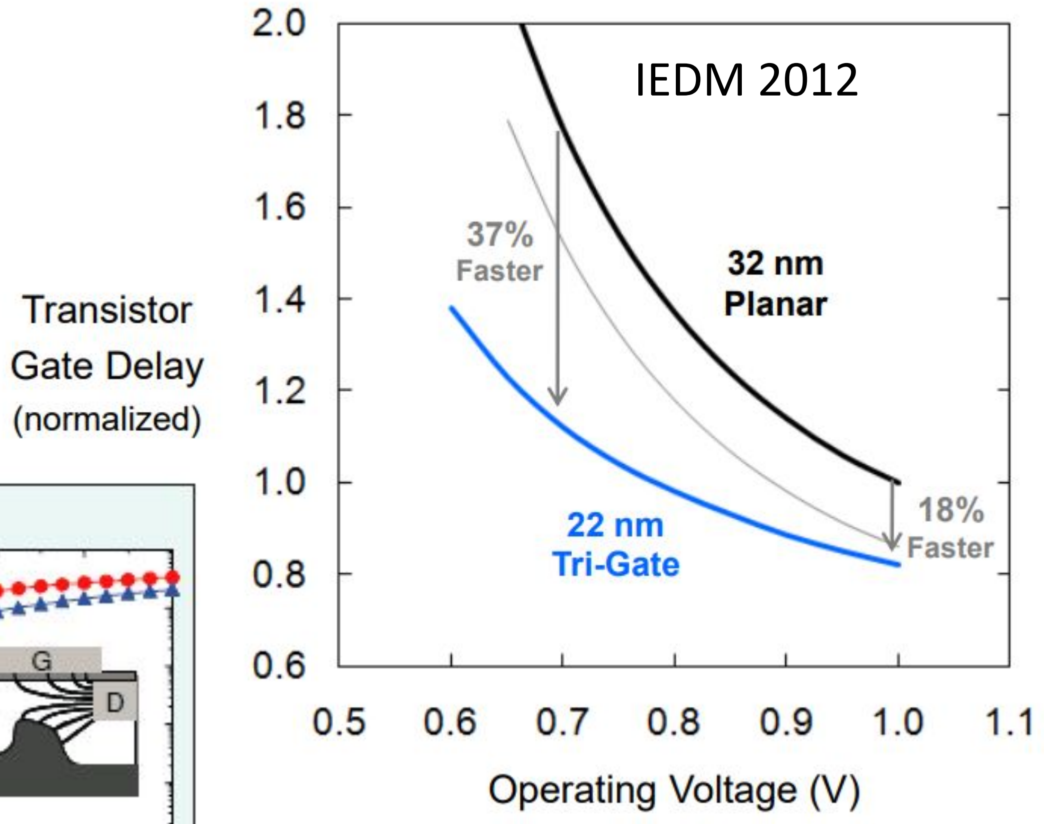
https://www.intel.com/content/dam/doc/technology-brief/32nm-logic-high-k-metal-gate-transistors-presentation.pdf

How does adding another gate help?



Double gate FETs have lower DIBL and SS compared to single gate FETs for the same gate length. This means it has higher I_{ON} for the same I_{OFF} or lower I_{OFF} for the same I_{ON}

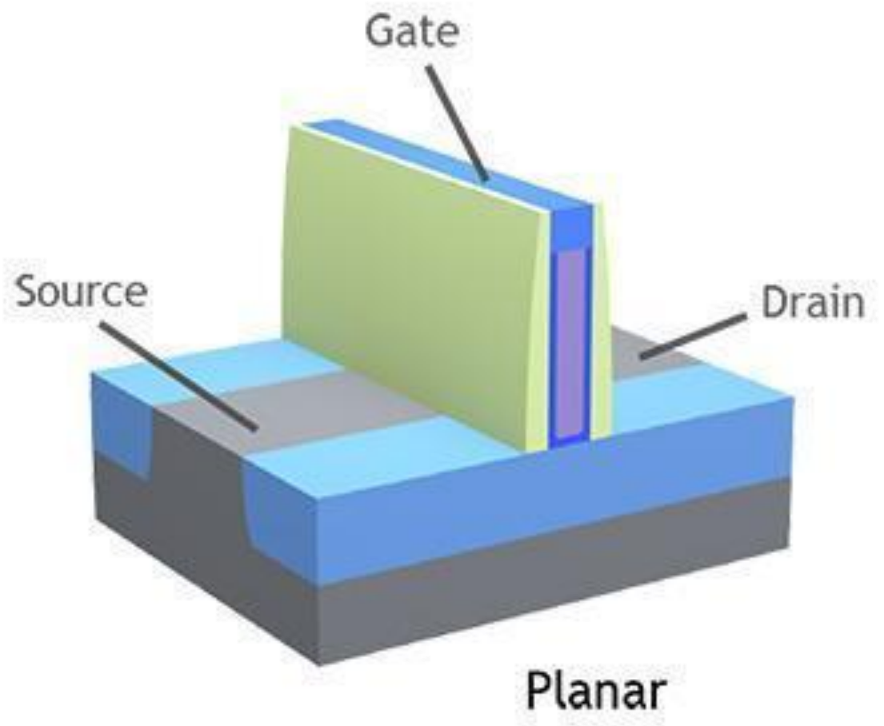


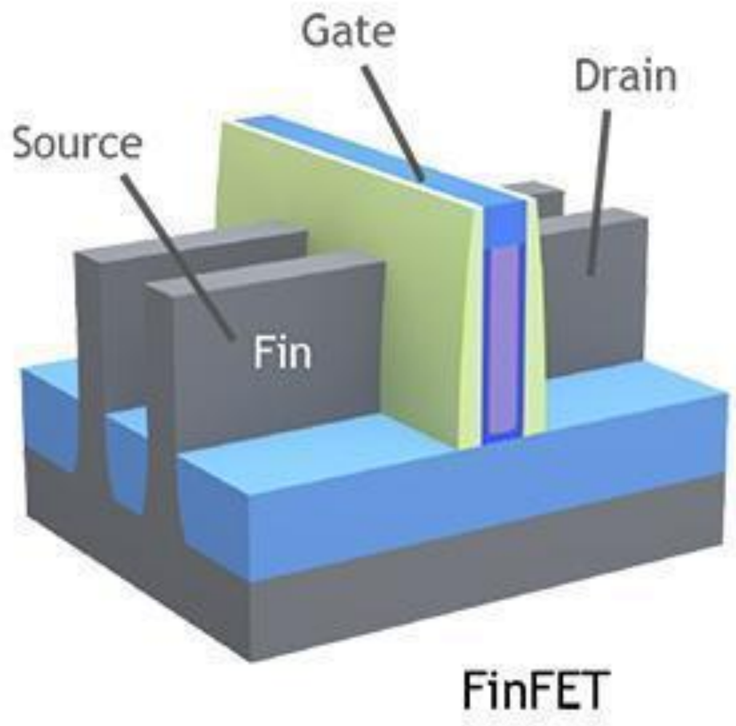


The performance improvement in speed increases at lower voltages

https://ieeexplore.ieee.org/document/1263404
https://download.intel.com/newsroom/kits/22nm/pdfs/22nm-Details_Presentation.pdf

How does a double gate FET looks like?





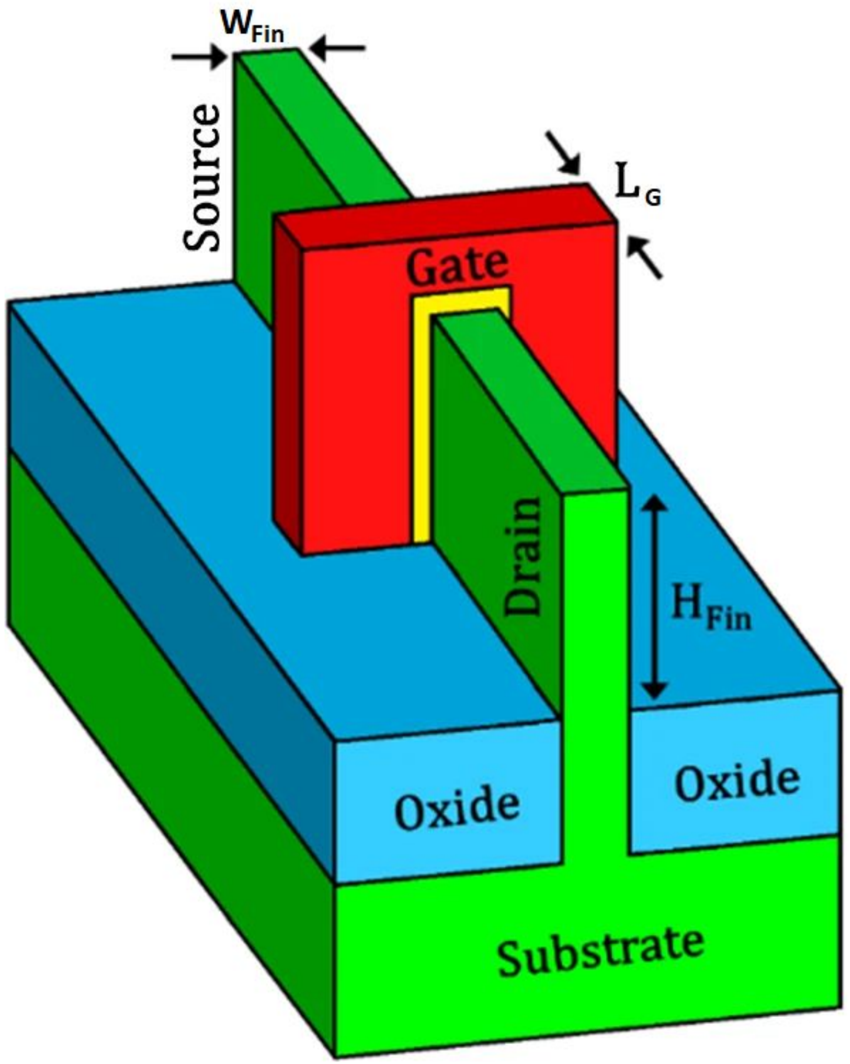
Single gate on the top of channel

One gate/channel is over front face of the fin, 2nd is over the back face of the fin and a 3rd (smaller than 1st and 2nd) on top of the Fin.

FinFET is the only manufacturable double gate solution. Gate/spacer is deposited & etched on all three sides of the fin in a single step. This ensures L_{Gate} and C_{ov} is same on all sides.

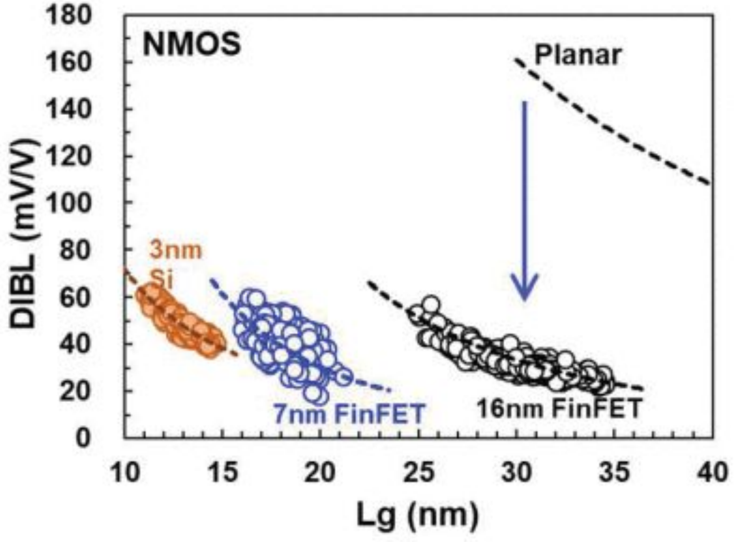
How is FinFET different from a planar FET?

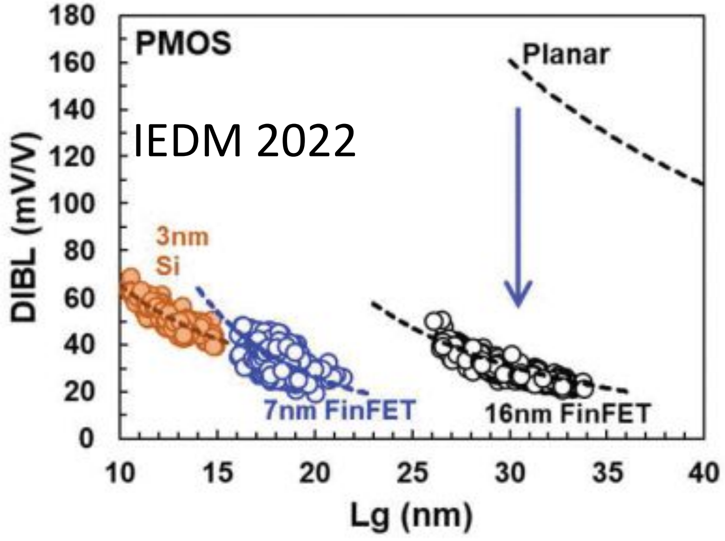
- Effective width of a FET with one fin=$2H_{FIN} + W_{FIN}$
- For more width, one has to add more fins. Allowable FET width comes in discrete numbers compared to continuous in planar FETs
- W_{FIN} should be less than $0.6L_G$ to have lower DIBL and SS compared to planar gate.
- Realizing fins smaller than gate needs more complex fabrication flow & double/quad patterning
- Front & Back surface of the fin has a different crystal orientation compared to top or that of the wafer
- Very little doping in the channel. Different threshold voltages are realized using gate work function engineering



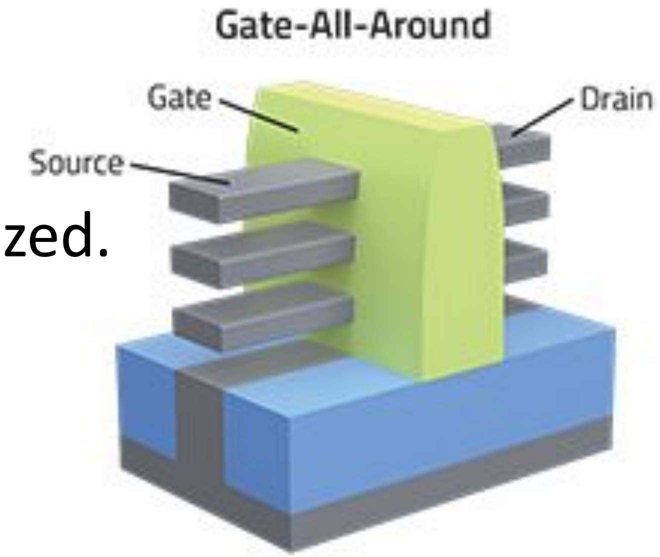
What comes after FinFET?

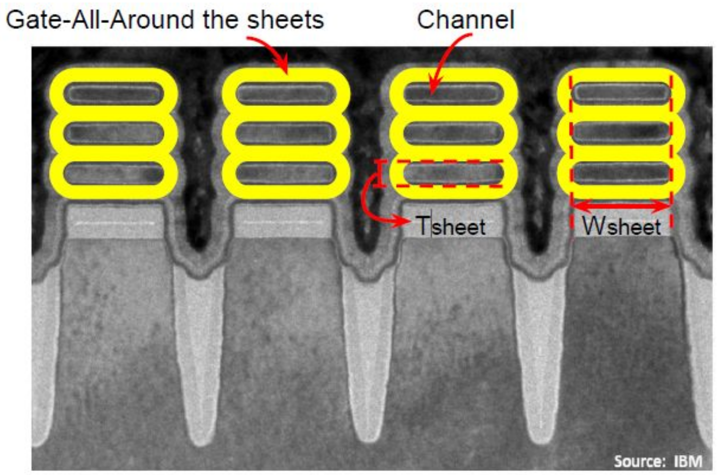
- FinFETs have been in production since 2011. TSMC 3nm node FET has a gate length > 10 nm
- Needs nanosheets of Si with gate wrapped around on all sides for gate length scaling below 10 nm





- Nanosheets have also better current/footprint than nano wires. Nanosheet width (W_{sheet}) is not quantized.
- More complex fabrication than FinFET





- 1st product - Samsung Exynos W1000 inside Galaxy Watch7 (2024)

https://newsroom.lamresearch.com/FinFETs-Give-Way-to-Gate-All-Around
https://spectrum.ieee.org/the-nanosheet-transistor-is-the-next-and-maybe-last-step-in-moores-law